Liking and Following and the Newsvendor: Operations and Marketing Policies under Social Influence

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We consider a monopolistic firm selling two substitutable products to a stream of sequential arrivals whose purchase decisions can be influenced by earlier purchases. Before demand realizes, the firm faces a newsvendor problem for the two products with economies of scale in production for each. When consumers are responsive to others' decisions, social influence amplifies demand uncertainty, leading to a lower profit for the firm. We propose three solutions for the firm to better cope with or even benefit from social influence: reduced product assortment, influencer recruitment, and production postponement. Firstly, as the potential substitutability between products increases due to social influence, the firm may leverage the increased substitutability and enjoy lower cost in production by reducing product assortment. Secondly, the firm can offer promotional incentives to recruit consumers as influencers. We reveal an operational benefit of influencer marketing that a very small fraction of such influencers is sufficient to diminish sales' unpredictability. Lastly, production postponement, allowing the firm to react to pre-order information, improves the accuracy of demand forecast. The optimal production time is highly sensitive to consumers' responsiveness to others' decisions when marginal cost increase from postponing production over time is moderate.

1. Introduction

It has long been recognized that consumer choice is influenced by the actions taken by others. It can be shaped by word of mouth, e.g., face-to-face recommendation by a friend, or by observational learning, e.g., by seeing what a stranger is wearing on the street. Thanks to advances in online shopping, the importance of social influence has never been greater. Online retailers place Facebook "Like" or Twitter "Share" buttons on their product pages where consumers can share their purchases with friends on Facebook or "followers" on Twitter simply by clicking those buttons. The count of all the likes or shares is updated in real time next to the buttons, and is readily observable by anyone who visits the webpage. Moreover, it has become increasingly common for

firms to collect sales information in real time and share it with consumers. For example, Amazon updates the sales ranking information of tens of thousands popular products hourly and presents it on products' webpages as part of the products' attributes.

Researchers have provided experimental and empirical evidence that social influence results in the phenomenon where "the rich get richer." Cai et al. (2009) report on a field experiment in a restaurant, observing an up-to-20% increase in demand for the five most popular dishes when consumers are given ranking information. With the prevalence of information regarding preferences and decisions of others, such a "preferential attachment" effect or "Matthew effect" (Merton 1968) can be even stronger in the online consumer market. Using the sales ranking on Apple's App Store, Carare (2012) shows that the willingness to pay of consumers is significantly higher for a top-ranked product than for the same product when it is unranked. Moreover, the collective choice of masses is also reported to be highly random due to social influence. An experimental study shows that with the exposure to others' choices, an increasing level of unpredictability of market success is observed (Salganik et al. 2006). When individuals tend to conform to others within a society or a social group, product adoption becomes *path-dependent*; the demand does not simply aggregate pre-existing individual preferences. For example, apparel fashion trends can occur essentially when random word-of-mouth takes over.

Even in the era of online shopping, the physical world still moves at a slow pace. The lead time for procurement and production is relatively long, so that most ordering decisions for short lifecycle products are made ahead of the sales season and can hardly be changed during the season. For example, lead times are often more than 3 months in the apparel and footwear industry, while the fashion seasons themselves are as short as 2-3 months.

Given the prevalence of social influence on consumption, we build a stylized model aiming to answer the following research questions: What are the effects of social influence on the policies of a newsvendor-type firm? If its impact is negative, how can the firm counteract it, or maybe benefit from it? If its impact is positive, how can the firm take advantage of it?

We consider a monopolistic firm selling two horizontally differentiated products in a market with a predictably stable size. Before the sales horizon, the firm faces a newsvendor ordering decision for the products. We assume economies of scale in production for each. During the sales season, consumers sequentially arrive and make purchase decisions that may be influenced by earlier adoptions. We show that consumers' responsiveness to others' purchase decisions is a key factor in shaping the distribution of aggregate sales. When consumers are relatively responsive to others' decisions, social influence amplifies demand uncertainty facing the firm. The demand uncertainty

manifests itself as both *unpredictability* and *inequality*: the firm is unsure which product will be popular, and when one product becomes popular, it may be extremely popular. We show that, if the newsvendor does not actively adapt its policies, the impact of social influence is non-positive.

We consider three responses for the firm, namely, a reduced product assortment, the use of firm-recruited influencers, and production postponement. Because of social influence, the potential substitutability between the products increases, as consumers are more inclined toward a popular product. We show that the firm may leverage the increased substitutability between products, and enjoy lower cost in production by offering fewer product varieties. This strategy is shown to be profitable when consumers are sufficiently sensitive to others' decisions.

In a strategy of influencer recruitment, the firm may randomly target a small fraction of consumers, designated influencers, and offer economic incentives, like price discounts or coupons, tailored to them. Through social influence, the firm can manipulate the collective decisions of the market by injecting influencers into the sequential sales process. We show that a small fraction of influencers are sufficient to diminish demand uncertainty and provide an operational benefit.

On the supply side, if the firm is able to delay production, we show that, similar to the strategy of ex-ante reduced product assortment, the firm may be better-off producing fewer product varieties after observing pre-order information. This strategy takes advantage of the way social influence results in demand herding, and does not require consumers to be highly sensitive to others' decisions. Furthermore, we show that the optimal production time depends greatly on the consumers' sensitivity to others' decisions, when marginal cost increase from postponing production over time is moderate.

Lastly, we extend our strategy of randomly-recruited influencers to a dynamic promotion framework, where the firm is able to dynamically vary the price of one product, depending on the market share and inventory level of the products. The result suggests that, under social influence, the pricing strategy follows a two-stage approach, where lower prices are set initially to attract the majority of early arrivals, and once the market share of the product is sufficiently large, the firm can profit from social influence by charging higher prices.

Literature Review

Our work is related to the literature on social influence and observational learning (see a recent work by Young (2009) on social learning and the references therein). In papers such as Banerjee (1992) and Bikhchandani et al. (1992), individuals sequentially make decisions, conditioning both on one's endowed private signal about the state of the world and on all predecessors' decisions. Despite some similarities, our work differs from the observational/social learning literature in the following two

aspects. First, we assume that consumers have complete knowledge on quality of the products. That is, the products belong to search goods, instead of experience goods (see Cabral 2000). Second, and more importantly, our focus is on the impact of social influence on the newsvendor and the firm's counteracting policies, while the literature on social learning is primarily focused on the social-influence process itself. For example, previous work considers whether the aggregation of the information from individual agents can lead to learning the true underlying state of the world.

In the social learning literature, there are Bayesian and non-Bayesian approaches to modeling how individuals are influenced by others. Similar to the non-Bayesian approach, we specify decision rules regarding how later arrivals are influenced by earlier adoptions. This approach allows us to simplify the learning process under social influence and focus on the firm's policies. It also enables us to capture some real-life phenomena that are difficult to model using a Bayesian approach. For example, the firm can recruit ambassadors who may be immune from social influence. It would be difficult to model their persistent disagreements from others when applying the Bayesian approach.

The demand model in our paper shares some similarities with the works on forecasting and inventory management of fashion goods where previous sales are assumed to stimulate future demand. Ravindran (1972) investigates the optimal stocking level and the optimal duration of the selling season in a single-period inventory model. Hartung (1973) extends Ravindran (1972) into a multi-period model but with a fixed duration of the selling season. In studying the impact of inventory cost on retail assortment breadth, van Ryzin and Mahajan (1999) consider a trend-following model where each customer makes the same choice. Petruzzi and Monahan (2003) consider the accessibility of the secondary market for fashion products, and study the optimal termination time where the firm shall shift to the secondary markets from the primary selling season. Unlike these models where demand is assumed to follow an exogenous distribution, we endogenize demand uncertainty by modeling consumers' adoption process as a path-dependent process, relate the demand uncertainty to the consumers' sensitivity to others' decisions, and design strategies to shape the aggregate demand by influencing consumers' decisions.

The study of consumer behavior and its impact on firms' operational decisions is gaining increasing interest (see, e.g., Shen and Su 2007, Netessine and Tang 2009). Forward-looking consumer behavior has been considered in a wide range of models, e.g., Aviv and Pazgal (2008), Su and Zhang (2008) and Liu and van Ryzin (2008). These papers focus on capturing individual consumer rationality, while the aggregate behavior of the population results from the summed action of one or multiple, homogenous consumer segments.

In contrast, our work belongs to a scant body of research on "social" operations management, where social interaction among individual decision makers or segments of individuals is explicitly modeled. Tereyağoğlu and Veeraraghavan (2012) address pricing and production decisions when consumers value uniqueness of the products to display their exclusivity and social status. Our work considers the opposite type of social behavior, where consumers value a product more if it is more popular in the population. Similar consumer behavior is studied in a queueing setting by Veeraraghavan and Debo (2009) and their subsequent works on herding in queues, where consumers infer service quality from the length of queues using a Bayesian approach. While they focus on the signaling effect of queue lengths with rational herding as a result, we study the impact of social-influence-induced herding on sales uncertainty and propose marketing and operational policies to react to them. While Papanastasiou et al. (2013) and Papanastasiou and Savva (2014) investigate the implications of sharing and learning opinions about product quality on a single-product monopolist firm's inventory and pricing decisions, we consider a two-product newsvendor problem and focus on the benefits of various strategies in mitigating the negative impact of social influence.

2. The Model and Analyses

2.1. The Setup

Consider a monopolistic firm selling two horizontally differentiated products to a market of heterogeneous consumers. The two products, labelled A and B, are located at the ends of a unit Hotelling line $[0, 1]$. Consumer preferences are distributed along the line. For instance, the two products may vary in their color or style, in which consumers have different preferences. We consider a market with a fixed market size M. Because we focus on how social influence affects consumers' choices between the firm's two products, the demand for each product does have uncertainty. This setup fits well to a market that has a relatively predictable consumer base with consumers differing in tastes; for example, in the well-known Sport Obermeyer case (Hammond and Raman 1994), the firm commits to produce 20,000 units of fashionable ski parkas in total for the season, which may imply a relatively predictable total market size. The firm must decide how many units to produce for each design/product. This is driven by consumer tastes and fashion trends generated under social influence. Instead of working with the absolute number of units to produce, we normalize the total market size to 1 and consider the market share of each product.

The firm faces a newsvendor problem for the two products, with economies of scale in production for each product. Without loss of generality, the products are assumed to have zero salvage value at the end of the sales season. In the event of stock-out, unmet demand is lost without additional stock-out penalty. The firm needs to determine the production quantity of the two products before

observing demand. For simplicity, the scale economy in production is in the form of a fixed cost of producing each of the two products, denoted as K, and a constant marginal supply cost c per unit of market share. The cost structure is assumed to be identical for both products. It is worthwhile noting that the case of no economies of scale (i.e., $K = 0$) is a special case of our model.

We consider a multi-period, dynamic process where consumers sequentially arrive one per time period. Since the market size is finite, the sales horizon has a finite length. Upon arrival, each consumer is able to observe the decisions made ahead of him. The firm knows the probability distribution of the consumer-intrinsic preferences. This can be achieved through comprehensive market research before launching products. However, the firm still faces the uncertainty of the demand realization because of the path-dependent nature of the sales process under social influence.

2.2. Demand Side

2.2.1. Individual's Decision Making. We analyze how consumers' purchase decisions are shaped by individual preference and social influence. Without loss of generality, we assume that consumers are uniformly distributed over the preference line segment with an individual's preference/type denoted by $\delta \in [0, 1]$. The utility of choosing a product is given by the sum of a constant base utility, a disutility measuring the distance between an individual's ideal location δ and the location of the product, and a social utility, less the price. Let r_i, $i = A, B$, denote the base utility of product i, which is assumed to be homogeneous across all consumers. The information accessible to consumers is the aggregate purchase decisions of his predecessors. A constant marginal disutility θ is incurred for a unit distance between consumer's ideal preference and the position of the product. Let us denote $\boldsymbol{x} = (x_A, x_B)$, where x_i is the fraction of one's predecessors who have purchased product i, $i = A, B$. The social utility for product i is specified as a power function x_i^α, with $\alpha \geq 1$. Assume that the prices for both products are identical, say p, which is the case for Sport Obermeyer. The utility for a consumer with type $\delta \in [0, 1]$ is

$$u_A(\boldsymbol{x}) = r_A - \theta\delta + x_A^\alpha - p, \qquad u_B(\boldsymbol{x}) = r_B - \theta(1-\delta) + x_B^\alpha - p, \tag{1}$$

The social utility term for product i, x_i^α is increasing in the fraction of consumers who have purchased product i, x_i. This form expresses the value of conformity to popular preferences in a society. For simplicity, we assume $r_A = r_B \equiv r$ in the following analysis, with obtained insights carried over to the case where r_A and r_B are not equal. Further, we normalize θ to 1.

Denote by $N_{i,t}$ the number of consumers who have purchased product i at the end of any period t. Let $\boldsymbol{N}_t = (N_{A,t}, N_{B,t})$. Assume the initial condition $\boldsymbol{N}_2 = (1, 1)$. That is, at the end of the 2-nd period, we assume one consumer purchased product A, and another purchased product B. This

assumption gives a symmetric initial start for both products before any social influence takes place. It avoids trivial solutions where all consumers prefer the same product. Let $\boldsymbol{X}_t = (X_{A,t}, X_{B,t})$, where $X_{i,t} = N_{i,t}/t$ denotes the fraction of consumers who have purchased product i at the end of period t. Given $\boldsymbol{X}_t = \boldsymbol{x}$, the consumer arriving at period $t+1$ will purchase product i if and only if $u_i(\boldsymbol{x}) \geq u_{-i}(\boldsymbol{x})$ and $u_i(\boldsymbol{x}) \geq 0$. Since δ is uniformly distributed, the probability that a consumer arriving at period $t+1$ will purchase product i is given by

$$f_i(\boldsymbol{x}) = P(u_i(\boldsymbol{x}) \geq u_{-i}(\boldsymbol{x}), u_i(\boldsymbol{x}) \geq 0) = \Pi_{[0,1]}\left[\min\left\{r - p + x_i^\alpha, \frac{x_i^\alpha - x_{-i}^\alpha + 1}{2}\right\}\right], \qquad i = A, B, \quad (2)$$

where $\Pi_{[0,1]}[a]$ is the projection of a onto the line segment $[0,1]$. That is, $\Pi_{[0,1]}[a] = 1$ if $a > 1$, $\Pi_{[0,1]}[a] = 0$ if $a < 0$, and otherwise $\Pi_{[0,1]}[a] = a$. Let us denote $\boldsymbol{f}(\boldsymbol{x}) = (f_A(\boldsymbol{x}), f_B(\boldsymbol{x}))$. The random sales process in our model is a generalized Pólya urn process (Hill et al. 1980, Arthur et al. 1987). We call $\boldsymbol{f}(\boldsymbol{x})$ the *updating probability function* that generates this stochastic process.

The case with no social influence is defined as when consumers make decisions independently and solely contingent on their intrinsic preferences. Consumers' utilities are the same as specified in Equation (1) except without the social utility term. That is, for a consumer of type $\delta \in [0,1]$, his utilities of purchasing product A and product B are given by $\tilde{u}_A = r - \delta - p$ and $\tilde{u}_B = r - (1-\delta) - p$, respectively. We use $\tilde{}$ to denote the set of notations without social influence. This case is equivalent to the scenario as $\alpha \to \infty$, and it serves as a benchmark in our comparison below.

When there exists no social influence, the probability that a random consumer arriving at period $t+1$ will purchase product i is given by

$$\tilde{f}_i = P(\tilde{u}_i \geq \tilde{u}_{-i}, \tilde{u}_i \geq 0) = \Pi_{\left[0,\frac{1}{2}\right]}\left[r - p\right], \qquad i = A, B, \quad (3)$$

which is independent of the purchase decisions of his predecessors. If $r - p \geq 1/2$, the market is completely covered, and each product covers one half of the market. Otherwise, each product acts like a local monopoly, and captures $r - p$ market share. We start our analysis under the assumption that the market is completely covered when there exists no social influence.

Assumption 1 (Complete Coverage without Social Influence). $r - p \geq 1/2$.

We first show that when the market is completely covered without social influence, it is also completely covered with social influence. This result is summarized in Lemma 1. That is, social influence will not have a negative impact on the total market size. Admittedly, we disregard the potential positive impact of social influence on the market size by assuming that the market is completely covered without social influence. The base model is more suitable in explaining the

product markets where social influence will affect which specific product a consumer chooses, rather than whether or not a consumer will enter the market. We extend our model to the case where the market is not completely covered in Section 4.1. In this extension, we show that social influence can *expand* the market size for a product, and this effect can be substantial so that the firm is better-off producing fewer product varieties.

Lemma 1 (Complete Market Coverage under Social Influence). *Under Assumption 1, the market is completely covered with social influence.*

Lemma 1 indicates that, under Assumption 1, a consumer will purchase one of the two products upon arrival. Thus, given that $\boldsymbol{X}_t = \boldsymbol{x}$, the probability that a consumer arriving at period $t+1$ will purchase product i is reduced to

$$f_i(x_i) = \frac{x_i^\alpha - (1-x_i)^\alpha + 1}{2}, \quad i = A, B. \tag{4}$$

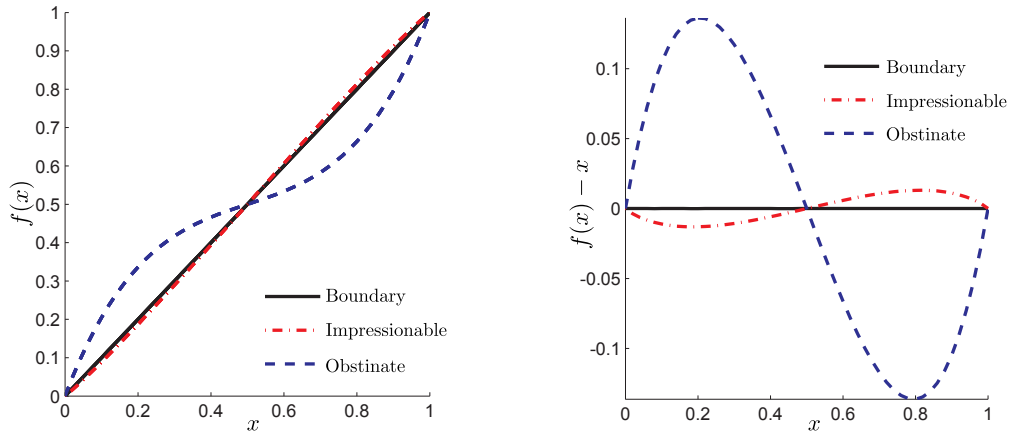
In our model, with Assumption 1 and a fixed market size, it suffices for us to focus discussion on the demand for one product, say product A. The demand for product B can be derived immediately. Thus we will omit the product subscript in the notation, i.e., $f(x)$ denotes the probability of a consumer choosing product A given that the fraction of his predecessors choosing product A is x.

2.2.2. Consumers' Sensitivity to Social Influence. The generated sales process can have dramatically different properties depending on the social utility exponent α.

Definition 1 (Consumers' Sensitivity to Social Influence). We define three scenarios based on the values of α: (1) Impressionable Scenario, when $1 < \alpha < 2$; (2) Boundary Scenario, when $\alpha = 1$ or 2; (3) Obstinate Scenario, when $\alpha > 2$.

Figure 1 shows the values of $f(x)$ and $f(x) - x$ with various α. First, we notice that $f(x)$ is always equal to $1/2$ when evaluated at $x = 1/2$, regardless of the value of α. This is consistent with the intuition that when the numbers of purchasers are equal for both products, social influence from early adopters of both products cancels out and consumers base their decisions solely upon their intrinsic preferences. Second, Figure 1 shows that the curve $f(x)$ becomes flatter within the small neighborhood around $x = 1/2$ as α increases. In this neighborhood, with $1 < \alpha < 2$, consumers respond actively to the difference between the market shares of the two products, namely, consumers are more likely to be lured towards the more popular product. However, as α increases, consumers become less responsive to social influence unless one product gains a dominant popularity. In the Boundary Scenario, the likelihood of purchasing a product is proportional to the observed market share. The case of $1 < \alpha < 2$ implies a stronger social influence than the Boundary Scenario, while

Figure 1 **The values of $f(x)$ and $f(x)-x$ as α varies.**



(a) $f(x)$ (b) $f(x)-x$

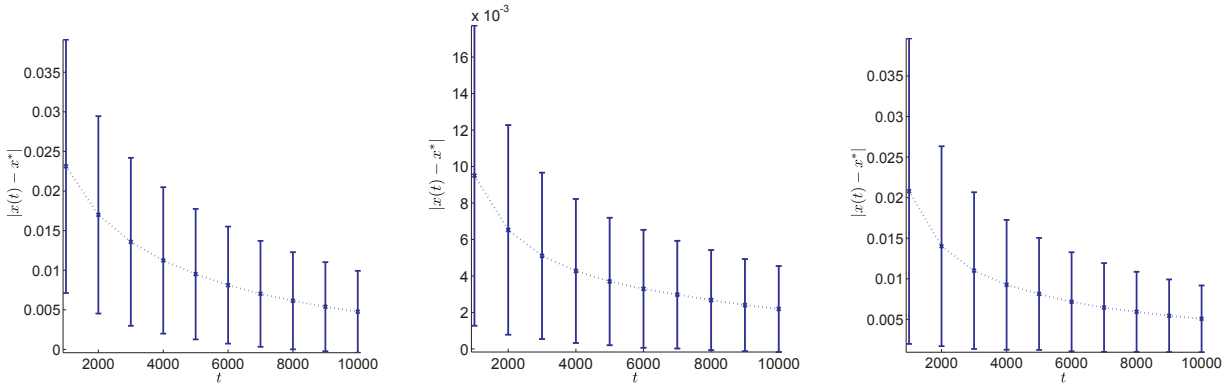
$\alpha > 2$ implies a weaker social influence than the Boundary Scenario. Thus, we interpret the power of the social-influence term, α, as the consumers' sensitivity with respect to others' decisions. It is easily seen, as an extreme case, that $\alpha \to \infty$ implies there is no social influence at all.

Most of the empirical literature on social influence assumes the boundary case where $f(x) = x$; see, e.g., Manski (1993). That is, individuals are proportionally influenced by the average decisions of their previous adopters. However, we will show shortly that the aggregate decisions of the entire population vary significantly across the three scenarios. By varying α, we span the whole spectrum of consumer sensitivities to social influence. Our definition of different scenarios can be taken as a categorization of product types. For some products, consumers are more likely to stick to their intrinsic preferences, while for the others, social influence plays a much more crucial role. Our analysis sheds some light on how a firm should respond to social influence based on the type of products it offers. We conclude that the specification of social utility, x^α, leads to intuitive outcomes supported by experimental evidences (e.g., Salganik et al. 2006). It is also consistent with rational herding theory from the economic perspective (e.g., Banerjee 1992, Bikhchandani et al. 1992) and conformism theory from the psychological perspective (e.g., Cialdini and Goldstein 2004).

2.2.3. Limiting Distribution of Aggregate Demand. The firm is particularly interested in the impact of social influence on the aggregate demand of each of the products. Because it is very cumbersome, if not impossible, to derive the distribution for a finite market, we resort to the limiting distribution of X_t, and use it as an approximation to the market share in our constant market size setting. We provide both theoretical and numerical support for this approximation.

First, we are able to analytically derive the convergence rate for some cases. For example, we show that if $f(x) = x$ has a single root, denoted by x^*, and $f'(x^*) < 1/2$, then the approximation error is $O(\sqrt{(\ln \ln t)/t})$ (see the proof of Lemma A1). Second, we conduct extensive numerical experiments to see how accurate the approximation is. Figure 2 illustrates the result with 5,000 sample paths for each market size. We observe that the approximation has an accuracy of 1-2% when the total market size is in thousands. The fast convergence to the limiting distribution indicates that the approximation becomes adequate when the market size is reasonably large.

Figure 2 **The absolute deviations of $x(t)$ from its corresponding convergence point x^* at various time periods t.**



(a) Impressionable Scenario ($\alpha = 1.5$) (b) Boundary Scenario ($\alpha = 2$) (c) Obstinate Scenario ($\alpha = 5$)

Note. The error bar is one standard deviation unit in length above and below the average absolute deviation of $x(t)$ from its corresponding convergence point x^*.

We next show that the asymptotic distribution of X_t depends solely on the updating probability function $f(x)$. The lemma below summarizes the limiting support for the distribution of X_t. It is a direct consequence of Hill et al. (1980, see, Corollary 3.1 and Theorem 4.2).

LEMMA 2 (ASYMPTOTIC DISTRIBUTION). *X_t converges almost surely to a random variable X^* with support contained in the set $\{x : f(x) = x, f'(x) \leq 1\}$.*

The points within the smaller set $\{x : f(x) = x, f'(x) < 1\}$ are referred to as the *stable equilibria* of the demand process. The basic idea of the convergence proof is as follows. The conditional expectation of X_{t+1}, given the state $X_t = x_t$, is $E[X_{t+1} \mid X_t = x_t] = x_t + \frac{1}{t+1} \cdot [f(x_t) - x_t]$. Thus the conditional mean of the increment $X_{t+1} - X_t$ shares the same sign as that of $f(x_t) - x_t$. After a finite number of periods, the process will converge to the points where the increment is zero, that

is, those solutions to the equation $f(x) = x$. The argument in Lemma 2 is similar to the *rational expectations equilibrium* (REE) that is often used in the economics and operations literature (see, e.g., Su and Zhang 2008). The condition $f'(x) < 1$ guarantees that the equilibrium is robust to small perturbations, and thus it is called the stability constraint. Our model of a random process is a dynamic view of how an REE is reached (Easley and Kleinberg 2010, Section 17.4). The limiting distribution of the sales process characterized by Equation (4) is summarized in the next lemma.

LEMMA 3 (LIMITING DISTRIBUTION OF AGGREGATE DEMAND). *The distribution of* $X^* \equiv \lim_{t\to\infty} X_t$ *is*

(i) Bernoulli with support on $\{0,1\}$ *and the success probability* $P(X^* = 1) = 1/2$ *under the Impressionable Scenario;*

(ii) uniform in $[0,1]$ *under the Boundary Scenario;*

(iii) the constant $1/2$ *under the Obstinate Scenario.*

Lemma 3 provides some justification to our definition of the three scenarios. Under the Impressionable Scenario ($1 < \alpha < 2$), consumers are easily influenced by others, and thus eventually only one of the two products crowds out the market with an equal chance. Under the Obstinate Scenario ($\alpha > 2$), the intrinsic preference dominates in one's decision making, and the aggregate demand converges to the average preference across the entire population as if consumers are not aware of others' decisions. In this case, even though social influence plays a role in an individual's decision making, it has no impact at the aggregate level in the limit. The Boundary Scenario is a compromise of the other two scenarios, which leads to a uniformly distributed limiting distribution.

Lemma 3 shows that the impact of social influence is two-fold: (1) On the (potentially) positive side, under the Impressionable Scenario it leads to a dominant popularity of one of the two products, and thus the firm may enjoy lower supply costs if there exist economies of scale in production. (2) On the negative side, under the Boundary and Impressionable Scenarios, social influence amplifies the demand uncertainty facing the firm, as the realized demand is path-dependent, making it harder to predict sales and decide production decisions ex ante.

2.3. Impact of Social Influence

In this section, we investigate the impact of social influence on a newsvendor's profitability through a detailed profit-cost analysis. Let X^* denote the random limit that the sales process converges to, i.e., the market share for product A in the limit. Denote by q_i, $i = A, B$, the firm's choice of production quantity for product i. Then, the newsvendor's expected profit can be written as

$$\pi(q_A, q_B) \equiv pE\Big[\min\{q_A, X^*\} + \min\{q_B, 1 - X^*\}\Big] - \Big[c(q_A + q_B) + 2K\Big]. \tag{5}$$

This is a formulation of the newsvendor problem for two products, where the first term expresses the total revenue from selling the two products, and the second term expresses the production cost.

By Equation (3) and Assumption 1, we know that the market share of each product is equal to 1/2 in the limit when there exists no social influence. Given the distribution of X^* under social influence as summarized in Lemma 3, we can derive the firm's profit with social influence, and compare it with the profit without social influence. The result is summarized as follows.

PROPOSITION 1 (IMPACT OF SOCIAL INFLUENCE). *(i) Under the Impressionable Scenario and the Boundary Scenario, the firm makes a lower profit with social influence;*

(ii) Under the Obstinate Scenario, the firm makes the same profit in both cases.

Proposition 1 shows that social influence never benefits the newsvendor. This implies that if a multi-product newsvendor already has a desired market coverage, the firm would want to curb social influence, e.g., by not posting sales information on its website. When consumers are sensitive to others' decisions, either one of the products can have greater sales. Because of the path-dependent nature of sales, there is greater demand uncertainty introduced by social influence. The demand uncertainty manifests itself as both unpredictability and inequality: a firm is unsure which product will be popular, and when one product becomes popular, it may be extremely popular. Consequently, if the newsvendor does not actively adapt its policies, this type of social influence can simply works against the firm.

3. Strategies under Social Influence

If social influence is inevitable, a newsvendor-type firm may suffer. How the firm should adapt its strategies? In addition, there could be a positive side of social influence: One of the products may become extremely popular, and thus the firm would be able to enjoy lower cost in production due to economies of scale. Is it possible for the firm to alleviate the demand uncertainty generated by social influence, while enjoying economies of scale in production at the same time? In this section, we propose three strategies to address these questions.

3.1. Reduced Product Assortment

With social influence, a consumer is more willing to tolerate the difference between the position of the product and his ideal preference if the product is popular in the population. That is, the potential substitutability between the products increases due to social influence. Consequently, the firm may leverage this increased substitutability among alternative options and enjoy lower cost in production, by offering fewer product varieties. In this section, we consider the case when the firm

reduces its product line from two products to one single product, and investigate when reduced product assortment may benefit the firm.

Suppose that the firm chooses to manufacture only product A (or equivalently, only product B). Similar to the base model, a consumer upon arrival is able to observe the fraction of his predecessors who have purchased product A. The utility of a δ-type consumer purchasing product A is similar to Equation (1). A consumer purchases the single product A as long as the utility is not less than 0. Given individuals' uniformly distributed preference, the probability that a random consumer arriving at period $t+1$ will purchase product A, when $X_t = x$, is $f(x) = \Pi_{[0,1]}\left[r - p + x^\alpha\right]$. Again, the consumer's purchase likelihood increases in the proportion of his predecessors who have purchased the product. In the context of producing only one product, Assumption 1 is no longer required. The market is not guaranteed to be covered completely with only one product even if $r - p \geq 1/2$ holds. Next, we derive the equilibrium that the process eventually converges to.

Lemma 4 (Aggregate Demand with One Product). *The aggregate demand X^* converges to the unique point of 1, if one of the following conditions holds:*

(i) (Value-Driven Regime) $r - p \geq 1$;

(ii) (Influence-Driven Regime) $r - p < 1$ *and* $\alpha \leq \bar{\alpha}$, *where* $\bar{\alpha}$ *is the unique solution of*

$$\alpha^{\frac{1}{1-\alpha}} - \alpha^{\frac{\alpha}{1-\alpha}} = r - p. \tag{6}$$

Otherwise, X^* follows a Bernoulli distribution with support on $\{x^*_\alpha, 1\}$, where x^*_α is the smaller root of $r - p + x^\alpha = x$.

Lemma 4 indicates that all consumers purchase product A in the limit, if either the intrinsic surplus of purchasing the product is sufficiently large, i.e., $r - p \geq 1$, or if consumers are sufficiently sensitive to the decisions of others, i.e., $\alpha \leq \bar{\alpha}$. The first condition can be derived from the standard Hotelling model without social influence, while the second condition reveals the impact of social influence. Even if the intrinsic surplus is not large enough to cover the entire Hotelling line segment, all consumers may still purchase the product in the limit due to the social utility. When the intrinsic surplus is small, i.e., $r - p < 1$, and consumers are not sensitive to others' decisions, i.e., $\alpha > \bar{\alpha}$, there exist multiple equilibria. The process may converge to 1, but it may also converge to an equilibrium where only x^*_α fraction of consumers purchase product A.

When producing only product A, the firm's expected profit is given by $\pi(q_A) = pE[\min\{q_A, X^*\}] - c \cdot q_A - K$. Under the conditions when the process is guaranteed to converge to 1 as shown in Lemma 4, the firm's optimal profit becomes $(p - c) - K$ with $q^*_A = 1$, which is

higher than the optimal profit that a firm can potentially obtain when producing both products, i.e., $(p-c)-2K$. Under those conditions, the firm is able to cover the entire market with a single product, and thus save the fixed cost that would be incurred when producing the second product.

Proposition 2 (Profit Comparison with Reduced Varieties). *The firm gains a higher profit by producing one product under either the value-driven regime or influence-driven regime.*

The benefit of producing only one product comes from two sources. On the demand side, the firm faces less uncertainty by producing one single product, and thus the firm can estimate demand more accurately ex ante, and plan production accordingly. On the supply side, the firm is able to enjoy economies of scale in production by saving the fixed cost of producing the other product.

Our view of a reduced product assortment can potentially explain the shrinking list of video games that was observed in the past few years (Wingfield 2013). In 2012, only half as many new games were released in American stores as in 2008. Video games exert strong social externality, especially in an online environment, where players are mostly attracted to the titles with the biggest pool of players. Though our model ignores competition, the benefits of the strategy may still be valid for a competitive market, not to mention that with less product varieties, the firms can have more budgets to create blockbusters. Our analysis suggests the need to carefully evaluate the impact of social influence on consumers' adoptions, and utilize the strategy of a reduced product assortment only when consumers are sufficiently sensitive to the decisions of others.

3.2. Influencer Recruitment

To improve profitability, the firm may be able to manipulate the collective decisions of the population by using social influence to its advantage. Consider the following model. Suppose the firm randomly targets a small fraction of consumers, say $\gamma \in (0,1]$, and offers them an economic incentive, like a price discount or coupon, to entice them to purchase a particular product, say product A. As social influence takes effect, we would expect that other consumers would be influenced by the decisions of these targeted consumers. Thus product A would become the more popular one within the entire market, allowing the firm to enjoy economies of scale in production *ex ante*.

We focus on the policy of *randomly* recruiting influencers, however, in practice, advances in information technology may allow firms to learn individual consumers' preferences with great accuracy, and hence can offer promotional incentives tailored to individual consumers. It is expected that with more *targeted* marketing campaigns, the influencers can be recruited more cost-efficiently. Thus our analysis is conservative in evaluating the strategy of influencer recruitment.

On another note, consumers are assumed to be homogeneous in our model in the sense that they are influenced by the aggregate sales. In reality, certain consumers, like celebrities, have an unparalleled impact on the population, and the firm may be better-off targeting them. Industry has adopted this strategy of influencer-marketing with some success (Fidelman 2013). However, the measurement of the success of such campaigns using sales data is still inadequate, if not non-existent. Our analysis illustrates the often-neglected operational benefit from recruiting influencers.

We call the targeted consumers *influencers*, and the remaining consumers *followers*. A randomly selected influencer, after being offered with the discounted price βp, where $\beta \in [0,1)$, would choose product A with probability

$$f_{I,A}(\boldsymbol{x}) = \Pi_{[0,1]}\left[\min\left\{r - \beta p + x_A^\alpha, \frac{x_A^\alpha - x_B^\alpha + 1 + (1-\beta)p}{2}\right\}\right],$$

and his probability of choosing product B is

$$f_{I,B}(\boldsymbol{x}) = \Pi_{[0,1]}\left[\min\left\{r - p + x_B^\alpha, \frac{x_B^\alpha - x_A^\alpha + 1 - (1-\beta)p}{2}\right\}\right],$$

where the subscript I denotes the choice probabilities of influencers. Given Assumption 1 and $\beta \in [0,1]$, the market is also completely covered in the existence of influencers. Thus, the choice probabilities can be simplified as $f_{I,A}(x) = \Pi_{[0,1]}\left[\frac{x^\alpha - (1-x)^\alpha + 1 + (1-\beta)p}{2}\right]$, and $f_{I,B}(x) = \Pi_{[0,1]}\left[\frac{(1-x)^\alpha - x^\alpha + 1 - (1-\beta)p}{2}\right]$. The choice probabilities of the followers remain given by Equation (4).

Under the scheme of randomly recruiting influencers, an arrival is either a follower with probability $1-\gamma$, or is recruited as an influencer with probability γ. Hence, the probability that an arrival prefers product A over product B is the weighted sum of the probabilities that a follower or an influencer chooses product A. Consequently, the updating probability function, given the state $X_t = x$, can be written as

$$f(x) = (1-\gamma)\frac{x^\alpha - (1-x)^\alpha + 1}{2} + \gamma\Pi_{[0,1]}\left[\frac{x^\alpha - (1-x)^\alpha + 1 + (1-\beta)p}{2}\right]. \tag{7}$$

Following our notation in the case when the market is completely covered, we omit product subscripts in the analysis. One question of particular interest is the determination of the fraction of influencers required to diminish demand uncertainty resulting from social influence. We show in the following proposition that in many cases it is sufficient to recruit only a *small* fraction of influencers.

Proposition 3 (Sufficient Conditions for Unique Convergence Point). *The set of convergence points of the sales process characterized by updating probability function (7) is a singleton if*

(i) $\gamma > 1 - \frac{2^{\alpha-1}}{\alpha}$ *and* $\gamma(1-\beta)p \geq \frac{2}{\alpha} - \frac{1}{2}$*, under the Impressionable Scenario. Moreover,* $1 - \frac{2^{\alpha-1}}{\alpha} \leq 5.8\%$ *when* $1 < \alpha < 2$*;*

(ii) $\gamma > 0$*, under the Boundary Scenario. Moreover, the singleton is 1.*

For the Obstinate Scenario, even without influencers, the sales process would converge to a deterministic outcome. For the Boundary Scenario, recall that the process converges to a uniform distribution when there are no influencers. Proposition 3(ii) implies that any *tiny* fraction of influencers ($\gamma > 0$) is sufficient to eliminate the demand uncertainty in the limit. Furthermore, under the Boundary Scenario, the updating probability function $f(x)$ is reduced to $x + \gamma \min\{1-x, (1-\beta)p/2\}$, whose *unique* stable equilibrium is 1. The influencers bias the collective adoption process so that the entire population purchase the same product in the limit. This result is *independent* of the fraction of influencers.

Under the Impressionable Scenario, each individual's decision is greatly influenced by the decisions from those who arrive before him and may have preferences in either direction. Our numerical experiments suggest that the Impressionable Scenario converges slower than the Boundary Scenario, and tend to imply that the Impressionable Scenario converges slightly slower than the Obstinate Scenario (see Figure 2). Thus, in the random sales process, a relatively larger fraction of influencers seems needed to eliminate demand uncertainty in the limit under the Impressionable Scenario than under the other two scenarios. Proposition 3 shows that even in this seemingly most unfavorable situation, a small fraction of influencers is sufficient: for $1 < \alpha < 2$, γ fraction of influencers is needed to diminish demand uncertainty, where $\gamma > 1 - \frac{2^{\alpha-1}}{\alpha}$. The maximum of $1 - \frac{2^{\alpha-1}}{\alpha}$ over $1 < \alpha < 2$ is $1 - e \ln 2/2 \approx 5.8\%$. That is, *at most* 5.8% of the demand must be influencers in order to completely eliminate demand uncertainty in the limit under the Impressionable Scenario.

The cost of recruiting influencers is quantified by the term $\gamma(1-\beta)p$, i.e., the product of the fraction of recruited influencers and the discount given to shape the decisions of these randomly targeted consumers. Proposition 3 indicates that the cost of recruiting influencers is negligible under the Boundary Scenario, i.e., the uncertainty in demand vanishes in the limit as long as $\gamma(1-\beta)p > 0$. However, under the Impressionable Scenario, the firm needs to spend a minimal cost of $(2/\alpha - 1/2)$ to eliminate demand uncertainty. Thus, when consumers are relatively sensitive to others' decisions, the strategy of recruited influencers is profitable if and only if the benefit from reduced demand uncertainty outweighs the cost of recruiting influencers.

3.3. Production Postponement

Production flexibility can be another solution to cope with the greater uncertainty induced by social influence. Flexibility in production technology enables firms to respond to the pre-order or

early sales information in a relatively short period of time. Production postponement can also be achieved through delayed differentiation. This policy of production postponement has been well studied in the operations literature in the context of demand uncertainty. For example, early sales information typically improves the accuracy of demand forecasting. The ability to quickly respond to early sales information can result in less overstocking and higher product availability (Fisher et al. 1994). In the context of capacity planning, the early sales information also helps the firm make a more informed capacity planning decision, and thus can significantly improve the firm's profit (Boyacı and Özer 2010). However, the value of production postponement has not been previously studied in the presence of social influence. Pre-order or early sales can be more than a good indicator of future market prospect; it also shapes purchasing decisions of later arrivals through social influence. As a result, the role of demand uncertainty under social influence can be significantly reduced, if the firm is able to postpone production until it observes sufficient pre-order or early sales information.

We seek to rigorously evaluate the value of production postponement for a newsvendor under social influence. Because we assume a fixed market size, the value of production postponement in our context is completely attributable to better reaction to social influence: It is easy to see that production postponement has zero benefits when there is no social influence. We first consider the case when the firm can delay production until after observing T periods of pre-order sales, for some given T. This would require a sufficiently flexible manufacturing system where the firm can produce in small batches, or allow consumers to pre-order products before they are actually produced. Such a pre-order strategy is gaining traction in the fashion industry. For example, Moda Operandi, a fashion online merchant, allows consumers to pre-order "looks" straight from the runway.

We assume Assumption 1 holds, i.e., the market is completely covered when there exists no social influence. Thus, the market is also completely covered under social influence. The profit function of the firm is the same as Equation (5), except that the distribution of X^* no longer follows the results in Lemma 3, given the pre-orders from the first T consumers. We summarize the updated distribution of X^* with pre-order information as follows. The Obstinate Scenario is trivial because the market shares of the two products can be accurately predicted in the limit as $1/2$ even without any pre-order information. The results under the Boundary and Impressionable Scenario directly follow from Zhu (2009, Theorem 2.1.4) and Auriol and Benaim (2000, Theorem 2), respectively.

Lemma 5 (Demand Distribution with Pre-order Information). *Given that $N_T = n < T$ where $T > 2$, the distribution of X^* is*

(i) a Bernoulli distribution with support $\{0,1\}$ *under the Impressionable Scenario. In particular, there exists a positive number* k *such that*

$$P(X^*=1)\begin{cases} \geq g(T,k) & \text{if } \frac{n}{T}>\frac{1}{2}, \\ \leq 1-g(T,k) & \text{otherwise,} \end{cases}$$

where $g(T,k)\equiv\left(1-\frac{2}{kT}\right)e^{-k/T}$*;*

(ii) a $Beta(n,T-n)$ *distribution under the Boundary Scenario;*

(iii) the constant $1/2$ *under the Obstinate Scenario.*

Lemma 5 implies that the level of demand uncertainty facing the firm decreases in the production delay under the Impressionable and Boundary Scenarios. Consider first the Impressionable Scenario. From Lemma 3, we see that without pre-order information, one of the two products will be extremely popular, and each of the two products has an equal chance of becoming the hit. With pre-order information, the firm would naturally expect that product A will become the popular one if the fraction of consumers purchasing product A at the end of period T, i.e., X_T, is greater than $1/2$, and would expect product B otherwise. The trend may still be reversed by the later arrivals, however, the likelihood of a reversal is no more than $1-g(T,k)$, which is decreasing in T. That is, uncertainty still exists for future sales due to the path-dependent nature of the process, however it is decreasing with more pre-order information. Similarly, under the Boundary Scenario, the variance of X^* is $\frac{n(T-n)}{T^2(T+1)}\leq\frac{1}{4(T+1)}$ (when $T>2$) which is less than, $1/12$, the variance of the uniform distribution on $[0,1]$.

3.3.1. Endogenized Product Assortment Decisions under Production Postponement. After collecting pre-order information, the firm may find it unprofitable to manufacture both products. In this subsection, we allow the firm to choose whether or not to produce the products after observing T periods of pre-orders, and investigate under what conditions the products may be dropped. We summarize the results in the lemma below.

Lemma 6 (Product Assortment under Production Postponement). *Given that* $N_T=n<T$ *where* $T>2$*,*

(i) under the Impressionable Scenario, if $n/T<1/2$*, the firm would choose not to produce product* A *if* $p(1-g(T,k))\leq c+K$*; if* $n/T\geq 1/2$*, the firm would not produce product* B *if* $p(1-g(T,k))\leq c+K$*;*

(ii) under the Boundary Scenario, the firm would choose not to produce product A *if and only if* $\frac{p\cdot n}{T}I_{I^{-1}_{1-c/p}(n,T-n)}(n+1,T-n)\leq K$*; and the firm would not produce product* B *if and only if* $\frac{p\cdot n}{T}[1-I_{I^{-1}_{c/p}(n,T-n)}(n+1,T-n)]\leq K$*, where* $I_x(\alpha,\beta)$ *and* $I_x^{-1}(\alpha,\beta)$ *denote the cumulative distribution function and inverse cumulative distribution function of* $Beta(\alpha,\beta)$*, respectively.*

As uncertainty for future demand decreases with pre-order information, the prediction for demand realization becomes more accurate. If the product is not likely to become popular, i.e., its market share is not likely to be large enough such that the profit from selling the product is able to cover the fixed production cost K, then the firm is better-off dropping the product from its product line. Compared with the strategy of reducing the product assortment ex ante, as studied in Section 3.1, the viability doing so under production postponement does not require deriving high surplus from the products (value-driven regime) or consumers' being highly sensitive to others' decisions (influence-driven regime). However, it does require a flexible manufacturing system such that products can be produced and delivered to markets quickly.

3.3.2. Endogenized Production Time. Next we allow the firm to choose when to produce, and investigate how the optimal production time depends on consumers' sensitivity to others' decisions. As postponing production means a shorter time to deliver products to market, the cost of production inevitably increases in the time delay for production. Essentially, the firm faces a trade-off between the reduced uncertainty in future demand and the increased production cost. The complexity of the demand distributions prevents us from identifying the structure of the optimal time to produce. However, we show in the example below that the optimal production time can be highly sensitive to consumers' sensitivity to others' decisions, i.e., α.

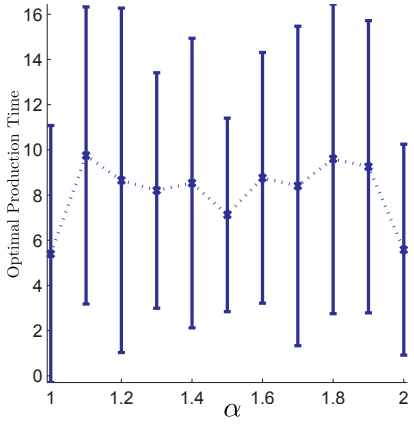
Example 1 (Sensitivity of Optimal Production Time). Let us denote the optimal production time as T^*_α, with a given α. For ease of presentation, we assume that only the fixed cost is increasing in the production delay, say $K_t = \tau_t K$, where τ_t is greater than or equal to 1, and non-decreasing in t. The expected profit when producing at time T, with production quantities q_A and q_B, is given by $\pi_T(q_A, q_B) = pE[\min\{q_A, X^*\} + \min\{q_B, 1 - X^*\}] - [c \cdot (q_A + q_B) + 2\tau_T K]$.

Note that under the Obstinate Scenario, the firm knows that demand will be split evenly between the two products even without any pre-order information. Thus, the firm does not benefit from postponing production, and the optimal production time under the Obstinate Scenario is to start production at the beginning of the sales season, i.e., $T^*_\alpha = 0$ for all $\alpha > 2$. Under the Boundary Scenario, given that $N_T = n$, we know that X^* follows a $Beta(n, T-n)$ distribution. The expected profit when the firm produces at any time $T \geq 0$ is bounded by $-2\tau_T K \leq \pi_T(q_A, q_B) \leq p - c - 2\tau_T K \leq p - c - 2K$, where the lower bound is derived when $q_A = q_B = 0$. The second inequality is shown in the proof of Proposition 1, and the third inequality is due to $\tau_t \geq 1$. As T goes to infinity, we know from Lemma 5 that the variance of X^* converges to 0, and the distribution degenerates to a single constant. Thus if marginal cost increase from postponing production, $\tau_{T+1} - \tau_T$, is sufficiently small, the optimal production time can be very large and the optimal profit is close to

$p-c-2K$. That is, under the Boundary Scenario, i.e., $\alpha=2$, the benefit of extra sales information can outweigh the marginal increase in production cost for a long time, leading to a large optimal production time. However, when α is slightly greater than 2 as under the Obstinate Scenario, the optimal decision is always to produce without postponement. Consequently, when the cost of postponing production increases slowly over time, T_α^* can vary significantly for a small change in α around 2.

On another note, the benefit from postponing production is bounded by $p-c$, because $-2\tau_T K \le \pi_T(q_A,q_B) \le p-c-2\tau_T K$ for all $T\ge 0$ and τ_T is non-decreasing in T. When τ_T increases quickly, say $\tau_{T+1}-\tau_T \ge (p-c)/K$ for some $\tilde{T}$, it will never be optimal for the firm to produce after period $\tilde{T}$. As a direct consequence, the firm produces without postponement when the preceding inequality is satisfied for $\tilde{T}=0$. The sensitivity of the optimal production time with respect to the consumers' responsiveness to others' decisions can be moderated when the marginal cost increase from production postponement is substantial.

Figure 3 **The optimal production time T_α^* as α varies.**



Note. The parameters are specified as follows. δ follows a uniform distribution with support $[0,1]$, and $p=10$, $c=3$, $K=1$, $\tau_t=0.04$. We run the simulation 1,000 times for each value of α. The error bar is one standard deviation unit in length above and below the average optimal production time T_α^*.

To further understand how the optimal production time depends on α, we conduct comprehensive numerical analysis under the Boundary and Impressionable Scenario, and the results are plotted in Figure 3. Our numerical results indicate that: (1) the optimal production time is generally smaller under the Boundary Scenario, i.e., $\alpha=1$ or 2, than that under the Impressionable Scenario, i.e., $1<\alpha<2$; (2) the optimal production time first tend to decrease, and then tend to increase as α varies within $(1,2)$. Notice that consumers are the most sensitive to others' decisions when α is

around 1.5. In that neighborhood, a small difference between the market shares of two products may influence the decisions of later arrivals substantially, and thus the firm can afford to produce as soon as a small lead in the market share by one product is established. However, as consumers become more independent, the trend is more likely to be reversed, and thus the firm needs to wait longer for more sales information before determining production quantities.

4. Extensions

We extend our model in two directions. First, we relax Assumption 1, and investigate how the impact of social influence changes when the market is not completely covered. Second, under the strategy of influencer recruitment, a small fraction of consumers are randomly selected and offered a pre-determined amount of economic incentives. In Section 4.2, we consider a dynamic framework, where the price of one product is determined based on a consumer's arrival time, as well as the market share and the inventory levels of the two products.

4.1. Incomplete Coverage

When $0 < r - p < 1/2$, the market is not completely covered when social influence does not exist. We ignore the case when $r - p \le 0$ because it is trivial – nobody purchases the products without social influence. The probability that a consumer arriving at period $t+1$ will purchase product i is

$$f_i(\boldsymbol{x}) = P(u_i(\boldsymbol{x}) \ge u_{-i}(\boldsymbol{x}), u_i(\boldsymbol{x}) \ge 0) = \Pi_{[0,1]}\left[\min\left\{r - p + x_i^\alpha, \frac{x_i^\alpha - x_{-i}^\alpha + 1}{2}\right\}\right], \qquad i = A, B,$$

given that $\boldsymbol{X}_t = \boldsymbol{x}$. Denote (X_A^*, X_B^*) the random limit the process converges to. The proposition below summarizes the limit when $0 < r - p < 1/2$.

Proposition 4 (Aggregate Demand under Incomplete Coverage). *Let x_α^* denote the smallest root of $r + x^\alpha - p = x$, if exists. Under the Boundary and Obstinate Scenario, the support of (X_A^*, X_B^*) is the same as that in Lemma 3 if and only if $r - p \ge 1/2 - (1/2)^\alpha$. Under the Impressionable Scenario, the support of (X_A^*, X_B^*) is the same as that in Lemma 3 if and only if $r - p \ge \alpha^{\frac{1}{1-\alpha}} - \alpha^{\frac{\alpha}{1-\alpha}}$. Otherwise,*

(i) under the Impressionable Scenario, (X_A^*, X_B^*) converges to either $(0,1)$, or $(1,0)$, or (x_α^*, x_α^*);

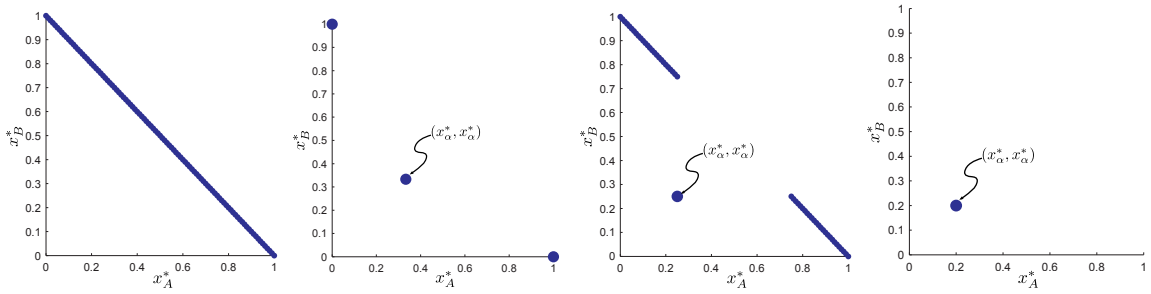
(ii) under the Boundary Scenario, when $\alpha = 2$, (X_A^*, X_B^*) converges to either $(x^*, 1-x^*)$, for some $x^* \in \left[0, \frac{1-\sqrt{1-4(r-p)}}{2}\right] \cup \left[\frac{1+\sqrt{1-4(r-p)}}{2}, 1\right]$, or (x_α^*, x_α^*);

(iii) under the Obstinate Scenario, the process converges to a unique equilibrium (x_α^*, x_α^*).

Potential equilibria when the market is not covered completely are illustrated in Figure 4. Under the Impressionable Scenario, besides the two equilibria where one product captures the entire market, another equilibrium where each product covers x_α^* market share emerges. Under the Boundary

Scenario, we need to differentiate between when $\alpha = 1$ and when $\alpha = 2$. The value of α affects the absolute value of a consumer's utility and thus whether or not a consumer purchases. When $\alpha = 1$, the market is completely covered and the process can converge to any point between 0 and 1. When $\alpha = 2$, the complete coverage is realized if and only if one product attracts the majority of the entire population, i.e., the market share of the popular product is greater than or equal to $\frac{1+\sqrt{1-4(r-p)}}{2}$, for any $r - p \in (0, 1/4)$. Under the Obstinate Scenario, instead of each product covering half of the entire market, each product only covers a smaller x^*_α share of the market.

Figure 4 An Illustration of Potential Equilibria when the Market is not Guaranteed to be Covered Completely.



(a) Boundary ($\alpha = 1$) (b) Impressionable (c) Boundary ($\alpha = 2$) (d) Obstinate

The result indicates a *positive* impact of social influence – an increase in the market size. Interestingly, the entire market will be fully covered under social influence, even if it is only partially covered without social influence. This happens when $r - p \in [1/2 - (1/2)^\alpha, 1/2)$ under the Boundary and Obstinate Scenario, and when $r - p \in [\alpha^{\frac{1}{1-\alpha}} - \alpha^{\frac{\alpha}{1-\alpha}}, 1/2)$ under the Impressionable Scenario. The combined market expansion from the two products can be as large as $(1/2)^{\alpha-1}$, which is substantial when α is small.

4.2. Dynamic Promotion

In this section, we consider a model where the firm can dynamically influence consumers' decisions by offering them economic incentives. The selling season has M time periods with one consumer arriving in each period. We index the time periods in reverse chronological order, with $t = 0$ representing the end of the selling season. In each period t, for every current inventory level $\boldsymbol{y} = (y_A, y_B)$ (or equivalently, a market share $\boldsymbol{x} = (x_A, x_B)$; see below), the firm needs to determine the current price of product A, which is denoted as p_t. The price of product B is assumed to be fixed at a base price point p, which can be arguably restrictive. However, as the demand of the products is mainly driven by the *difference* of the two products' prices, the optimal pricing of product A

sheds light on how the two products should be priced relatively given different market conditions and inventory levels.

Like revenue management problems, we assume that the initial production quantity of product i is exogenously given, denoted by q_i, $i = A, B$. Given inventory level $\boldsymbol{y}$ in period t, the market share of product i, $i = A, B$, can be derived as $x_i = (q_i - y_i)/(M - t)$. For ease of notation, we use the inventory level vector $\boldsymbol{y}$ and the market share $\boldsymbol{x}$ vector interchangeably in the following analysis. Let $V_t(\boldsymbol{y})$ denote the optimal expected revenue from period t to the end of the season. We can formulate the dynamic promotion problem as follows

$$V_t(\boldsymbol{y}) = \max_{p_t} \Big\{ f_A(p_t, \boldsymbol{y}) \Big(p_t + V_{t-1}(\boldsymbol{y} - \boldsymbol{e}_A) \Big) + f_B(p_t, \boldsymbol{y}) \Big(p + V_{t-1}(\boldsymbol{y} - \boldsymbol{e}_B) \Big) + \Big(1 - f_A(p_t, \boldsymbol{y}) - f_B(p_t, \boldsymbol{y}) \Big) V_{t-1}(\boldsymbol{y}) \Big\}, \tag{8}$$

where $f_i(p_t, \boldsymbol{y})$ denotes the probability that a consumer chooses product i in period t. The boundary conditions are given by $V_t(\mathbf{0}) = 0$ and $V_0(\boldsymbol{y}) = 0$. The consumers' utilities from purchasing the products are similar to those given by Equation (1), except that the price p_t of product A is dynamically determined by the firm. The choice probabilities, $f_A(p_t, \boldsymbol{y})$ and $f_B(p_t, \boldsymbol{y})$, are given by

$$f_A(p_t, \boldsymbol{y}) = \Pi_{[0,1]} \left[\min \left\{ r - p_t + x_A^\alpha, \frac{x_A^\alpha - x_B^\alpha + 1 + (p - p_t)}{2} \right\} \right],$$
$$f_B(p_t, \boldsymbol{y}) = \Pi_{[0,1]} \left[\min \left\{ r - p + x_B^\alpha, \frac{x_B^\alpha - x_A^\alpha + 1 - (p - p_t)}{2} \right\} \right],$$

where $x_i = (q_i - y_i)/(M - t)$, $i = A, B$.

Let $\Delta_t V_t(\boldsymbol{y}) \equiv V_t(\boldsymbol{y}) - V_{t-1}(\boldsymbol{y})$, $t = 1, \ldots, M$, and $\Delta_i V_t(\boldsymbol{y}) \equiv V_t(\boldsymbol{y}) - V_t(\boldsymbol{y} - \boldsymbol{e}_i)$, $i = A, B$. Define $g_t(p_t, \boldsymbol{y}) \equiv f_A(p_t, \boldsymbol{y})(p_t - \Delta_A V_{t-1}(\boldsymbol{y})) + f_B(p_t, \boldsymbol{y})(p - \Delta_B V_{t-1}(\boldsymbol{y}))$, then the optimality equation can be rewritten as $\Delta_t V_t(\boldsymbol{y}) = \max_{p_t} \{ g_t(p_t, \boldsymbol{y}) \}$. The optimal price p_t^* given inventory level $\boldsymbol{y}$ at time t is the one that maximizes $g_t(p_t, \boldsymbol{y})$, and the structure of p_t^* is summarized in the following lemma.

Lemma 7 (Optimal Dynamic Pricing). *Let us denote* $p_c^* \equiv p + \frac{1}{2} \Big(\Delta_A V_{t-1}(\boldsymbol{y}) - \Delta_B V_{t-1}(\boldsymbol{y}) + x_A^\alpha - x_B^\alpha + 1 \Big)$ *and* $p_m^* \equiv \frac{1}{2} \Big(\Delta_A V_{t-1}(\boldsymbol{y}) + x_A^\alpha + r \Big)$. *An optimal solution to Equation (8) is given by*

(i) When $x_B \geq (1 - r + p)^{\frac{1}{\alpha}}$, $p_t^* = \Pi_{[p_{\min}^c, p_{\max}^c]}[p_c^*]$;

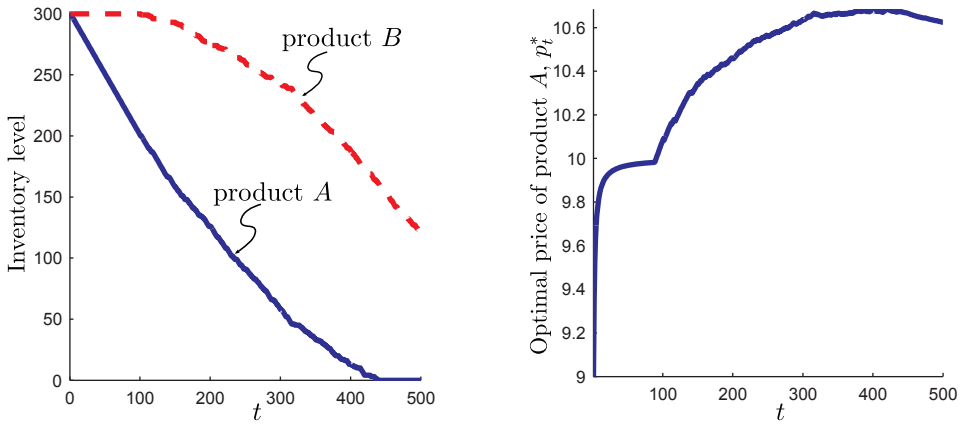
(ii) When $x_B < (1 - r + p)^{\frac{1}{\alpha}}$, $p_t^* = \Pi_{[p_{\min}^c, \tilde{p}]}[p_c^*]$ *or* $\Pi_{[\tilde{p}, p_{\max}^m]}[p_m^*]$,

where $p_{\max}^m \equiv r + x_A^\alpha$, $p_{\min}^c \equiv p + x_A^\alpha - x_B^\alpha - 1$, $p_{\max}^c \equiv p + x_A^\alpha - x_B^\alpha + 1$ *and* $\tilde{p} \equiv 2r + x_A^\alpha + x_B^\alpha - 1 - p$.

When $x_B \geq (1 - r + p)^{\frac{1}{\alpha}}$, irrespective of the price of product A, the entire market can be covered by product B. In this case, if it is optimal to price product A such that both products share the coverage of the entire market, then the optimal price is given by p_c^*. Otherwise, the optimal

price is given by either $p^c_{\min}$, which is the highest price the firm could charge for product A with the entire market being covered when both products A and B are available, or $p^c_{\max}$, which is the lowest price to shut down demand for product A when both products are available. The relative value of p^*_c compared with the price of product B, p, depends on the market shares and the marginal values of the products. The firm would charge a higher price for product A if and only if $V_{t-1}(\boldsymbol{y}) - \Delta_B V_{t-1}(\boldsymbol{y}) + x_A^\alpha - x_B^\alpha + 1 \geq 0$. One sufficient condition would be that product A has a greater market share, and the marginal value of having one extra unit of product A is greater than the value of having one extra unit of product B, i.e., $\Delta_A V_{t-1}(\boldsymbol{y}) \geq \Delta_B V_{t-1}(\boldsymbol{y})$. When $x_B < (1 - r + p)^{\frac{1}{\alpha}}$, an optimal price can arise from the local monopoly case, where there exists no overlap in the markets covered by the two products, or from the cannibalization case, where a portion of the market is covered by both products. The firm needs to compare the optimal profit derived from these two cases and choose the price that yields a higher profit.

Figure 5 An Example of Optimal Dynamic Promotion.



(a) Inventory Level (b) p^*_t

Note. The parameters are specified as follows. δ follows a uniform distribution with support $[0, 1]$, and $p = 10$, $r = 10.7$, $\alpha = 1.5$. The inventory levels at the beginning of the selling season are $q_A = 300$, and $q_B = 300$.

We also conduct numerical analysis to examine the structure of optimal dynamic promotion, and one typical sample path is shown in Figure 5. The result suggests a two-stage pricing strategy: (1) in the early stage, the firm would charge lower prices for product A than the base price point p such that the majority of early arrivals are lured towards product A; (2) once the market share of product A is sufficiently large, the firm then can profit from social influence by charging higher prices than the base price point p. Even if the price of product A is sometimes significantly higher

than that of product B, a large proportion of arrivals can still choose product A due to a high social utility. Moreover, the sample path of the inventory level for product A seems to closely follow a linear trajectory. This suggests that a feedback-type heuristic policy can be near optimal. The heuristic policy would repeatedly re-solve the deterministic control problem corresponding to the stochastic dynamic programming (see, e.g., Maglaras and Meissner 2006, Gallego and Hu 2014).

5. Conclusion

In this paper, we study the impact of social influence on consumers' purchase decisions, and subsequently the impact on the newsvendor's decisions. Social influence is a double-edged sword. The impact of social influence is two-fold: (1) on the positive side, the total or individual product market size increases due to an increased social utility; (2) on the negative side, the demand faced by the firm is more uncertain. If the firm already has a desired coverage in the absence of social influence, the negative side of social influence would dominate, resulting in a lower profit for the firm. We propose three solutions for the firm: reduced product assortment, influencer recruitment and production postponement. Taking advantage of increased substitutability among different products due to social influence, the firm can lower cost in production by reducing product assortment. This strategy is profitable for the firm when either the surplus provided by the product is sufficiently large, or consumers are sufficiently sensitive to others' decisions. Alternatively, the firm can manipulate the decisions of the entire population by introducing influencers. The operational benefit of an influencer strategy is that a *tiny* fraction of influencers is sufficient to diminish the demand uncertainty introduced by social influence. On the supply side, the strategy of postponing production allows the firm to improve the accuracy of demand forecast through pre-orders, but, the firm is faced with higher production cost at the same time. The optimal production time is highly sensitive to consumers' sensitivity to others' decisions when the marginal cost increase from postponing production over time is moderate.

Our proposed model provides an *endogenous* way to capture how fashion trends proliferate. The multi-period framework can be used to study many operational problems that previously considered "fashion" as exogenous demand uncertainty. For example, the model can be used to study dynamic pricing by competing firms when there is social influence. Moreover, the model may be extended to the case where there is market size uncertainty. Here a Bayesian model may be used in which a newsvendor can simultaneously learn about the market size and respond to social influence through delayed production decisions.

APPENDIX A: Rate of Convergence

Here we study how fast the sales process converges with influencers. In practice, because of the finite market size, the firm can only recruit a finite number of influencers by running promotions. The rate of convergence lends credibility to the accuracy of our approximation. Again for tractability, we focus on the cases when a sufficient condition in Proposition 3 is satisfied for a given scenario. The definition below formalizes the concept of the convergence rate utilized in this paper.

Definition 2 (Converge Faster). Let $\{Y_t\}$ and $\{Z_t\}$ be two discrete-time stochastic processes. Assume they converge to unique points, and let y^* and z^* be their unique convergence points, respectively. Suppose, for any discrete-time series $\{\beta_t\}$, $\limsup_{t\to\infty}\beta_t|Y_t - y^*| = \mu_y$ and $\limsup_{t\to\infty}\beta_t|Z_t - z^*| = \mu_z$. If $\mu_y < \mu_z$ ($\mu_y > \mu_z$), then we say that $\{Y_t\}$ converges at a faster (slower) rate than $\{Z_t\}$.

We can interpret this definition in the following sense. The limit superiors on $\{Y_t\}$ and $\{Z_t\}$ indicate that, for any $\epsilon > 0$, there exists T, such that $|Y_t - y^*| < (\mu_y + \epsilon)/\beta_t$ and $|Z_t - z^*| < (\mu_z + \epsilon)/\beta_t$ for all $t > T$. That is, after a finite number of observations, the process $\{Y_t\}$ is constrained within a narrower band around its convergence point than that of $\{Z_t\}$, if $\mu_y < \mu_z$. As a result, we say that $\{Y_t\}$ converges at a faster rate than $\{Z_t\}$.

Given the definition above, we compare the convergence rates under the policy of recruiting influencers to the case where all consumers' decisions are solely contingent on their own intrinsic preferences, that is, without social influence. Recall that the probability that any influencer chooses product A without social influence is $\Pi_{[0,1]}\left[\frac{1+(1-\beta)p}{2}\right]$, and the probability that any follower chooses product A without social influence is $1/2$. This process with independent decision making converges to $(1-\gamma)\frac{1}{2} + \gamma\Pi_{[0,1]}\left[\frac{1+(1-\beta)p}{2}\right]$ at an *exponential* rate given by the law of large numbers (Baum et al. 1962). Against this benchmark, we compare the case when consumers are susceptible to others' decisions. The result is summarized as follows.

Lemma A1 (Comparison of Convergence Rates). *Assume the firm recruits influencers such that the process converges to a unique stable equilibrium x^*. The process with social influence converges faster than the process without if and only if $\frac{x^*(1-x^*)}{f(1/2)(1-f(1/2))} < 1 - 2f'(x^*)$ and $f'(x^*) < 1/2$.*

Lemma A1 shows that social influence facilitates a marketing campaign in shaping a collective decision. This is because the convergence point is being pulled closer towards 1 with social influence than without social influence.[1] Notice that the rate of convergence is affected by the convergence

[1] Notice that the convergence point without social influence is $f(1/2) > 1/2$. Because of $f'(x) > 0, \forall\, x \in [0,1]$, we have $f(f(1/2)) > f(1/2)$. Thus the convergence point with social influence, x^*, which is given by $f(x) = x$, must be greater than $f(1/2)$.

point: ceteris paribus, the process converges in the slowest pace when the convergent point is 1/2, and the rate of convergence increases as the equilibrium deviates from 1/2.[2] When the unique stable equilibrium x^* is closer to 1, the condition $\frac{x^*(1-x^*)}{f(1/2)(1-f(1/2))} < 1 - 2f'(x^*)$ is more likely to hold, and consequently, we are more confident in using the limit as an approximation to our finite market-size model because of its fast convergence.

APPENDIX B: Proofs

Proof of Lemma 1. Recall that the probability that a consumer arriving at period $t+1$ will purchase product i, given $\boldsymbol{X}_t = \boldsymbol{x}$, is

$$f_i(\boldsymbol{x}) = P(u_i(\boldsymbol{x}) \geq u_{-i}(\boldsymbol{x}), u_i(\boldsymbol{x}) \geq 0) = \Pi_{[0,1]}\left[\min\left\{r - p + x_i^\alpha, \frac{x_i^\alpha - x_{-i}^\alpha + 1}{2}\right\}\right], \qquad i = A, B.$$

In order to show that the market is completely covered, it suffices to show that $r - p + x_i^\alpha \geq \frac{x_i^\alpha - x_{-i}^\alpha + 1}{2}$, $\forall\, x_i \in [0,1]$. As long as this inequality is satisfied, the non-purchase option never kicks in, and thus the market is completely covered. We have

$$x_i^\alpha + x_{-i}^\alpha \geq 0 \Leftrightarrow -x_{-i}^\alpha \leq x_i^\alpha \Leftrightarrow \frac{x_i^\alpha - x_{-i}^\alpha + 1}{2} \leq x_i^\alpha + \frac{1}{2} \Leftrightarrow \frac{x_i^\alpha - x_{-i}^\alpha + 1}{2} \leq r - p + x_i^\alpha,$$

where the first inequality is due to $x_i \in [0,1]$, and the last inequality is due to Assumption 1. Thus, we obtain the announced result. □

Proof of Lemma 3. When $\alpha = 1$ or 2, the updating probability function is reduced to $f(x) = x$. It immediately follows from Zhu (2009, Theorem 2.1.4) that X^* is uniformly distributed. When $\alpha \neq 1, 2$, it is easily verified that $f(x)$ has at least three fixed points within $[0,1]$, namely $0, \frac{1}{2}$ and 1. Next, we show that $f(x)$ does not have other fixed points. The second order derivative of Equation (4) is given by $f''(x) = \frac{(1-\gamma)\alpha(\alpha-1)}{2}\left[x^{\alpha-2} - (1-x)^{\alpha-2}\right]$. When $1 < \alpha < 2$, $f(x)$ is convex when $x \in [0, \frac{1}{2}]$ and concave when $x \in [\frac{1}{2}, 1]$. Let $g(x) = f(x) - x$. As a linear combination of a convex function and a linear function, $g(x)$ is convex as well when $x \in [0, \frac{1}{2}]$. The maximum of $g(x)$ over the interval $[0, \frac{1}{2}]$ is reached at end points, i.e., either 0 or $\frac{1}{2}$, which is $g(0) = g(\frac{1}{2}) = 0$, leading to $f(x) < x$ when $x \in (0, \frac{1}{2})$. Similarly, we have $f(x) > x$ when $x \in (\frac{1}{2}, 1)$. Consequently, when $1 < \alpha < 2$, $f(x)$ has no fixed points other than $0, \frac{1}{2}$ and 1. It is easy to verify that $f'(0) = f'(1) < 1$ and $f'(\frac{1}{2}) > 1$, and thus X^* has the support of $\{0, 1\}$. As $X_2 = \frac{1}{2}$, by symmetry, the process converges to either 0 or 1 with the same probability $\frac{1}{2}$. Similarly, we can show that $\frac{1}{2}$ constitutes the unique stable equilibrium when $\alpha > 2$. □

[2] To see this, consider an independent and identically distributed sequence of Bernoulli random variables with parameter $p \in [0,1]$. When $p = 0$ or 1, the average process of the sequence reaches its convergence point at either 0 or 1 at the first step and stays there forever. However when $p = 1/2$, the average process fluctuates, but eventually it converges to 1/2 by the law of large numbers.

Proof of Proposition 1. When there exists no social influence, we know that the market share of each product converges to $1/2$ in the limit. Thus the firm obtains the highest profit ordering $q_A = q_B = 1/2$, with an optimal profit of $p - c - 2K$.

Next consider the case with social influence. Under the Obstinate Scenario, the market share of product A, X^*, also converges to $1/2$ in the limit, as if consumers are not influenced by the decisions of others'. Thus, the firm makes the same profit as that in the case with no social influence.

Under the Boundary Scenario and the Impressionable Scenario, the aggregate demand X^* is a random variable as shown in Lemma 3, and we will show that the firm's profit given by Equation (5) is always no greater than $p - c - 2K$. Consider first the case when $q_A + q_B > 1$. Then we have $pE\Big[\min\{q_A, X^*\} + \min\{q_B, 1 - X^*\}\Big] - \Big[c(q_A + q_B) + 2K\Big] \le pE\Big[X^* + (1 - X^*)\Big] - c - 2K = p - c - 2K$, where the inequality is due to the first-order stochastic dominance of $X^* + (1 - X^*)$ over $\min\{q_A, X^*\} + \min\{q_B, 1 - X^*\}$, and $q_A + q_B > 1$.

Next consider the case when $q_A + q_B \le 1$. We have $pE\Big[\min\{q_A, X^*\} + \min\{q_B, 1 - X^*\}\Big] - \Big[c(q_A + q_B) + 2K\Big] \le (p - c)(q_A + q_B) - 2K \le p - c - 2K$, where the first inequality is due to the first-order stochastic dominance of $q_A + q_B$ over $\min\{q_A, X^*\} + \min\{q_B, 1 - X^*\}$, and the second inequality is due to $q_A + q_B \le 1$. Thus, we obtain the announced result. □

Proof of Lemma 4. We first show that the process converges to the unique point of 1, if $r - p \ge 1$, or $r - p < 1$ and $\alpha \ge \bar{\alpha}$. Recall that the probability that a random consumer arriving at period $t+1$ will purchase product A, given that $X_t = x$, is $f(x) = \Pi_{[0,1]}[r - p + x^\alpha]$. When $r - p \ge 1$, $f(x)$ is reduced to $f(x) = 1$, $\forall\, x \in [0, 1]$. Based upon Lemma 2, we can verify that 1 constitutes the unique equilibrium .

Next consider the case when $r - p < 1$. Let $\bar{x}_\alpha = [1 - (r - p)]^{1/\alpha}$, where $r - p + x^\alpha = 1$ when evaluated at $x = \bar{x}_\alpha$. Then, we can re-write $f(x)$ as

$$f(x) = \begin{cases} r - p + x^\alpha, & \text{if } 0 \le x < \bar{x}_\alpha, \\ 1, & \text{if } x \ge \bar{x}_\alpha. \end{cases}$$

Due to $r - p < 1$ and Assumption 1, we know that $\bar{x}_\alpha < 1$, and thus 1 is always an equilibrium. To this end, we need to show that there exists no other equilibrium when $\alpha \le \bar{\alpha}$. Let $g(x) = r - p + x^\alpha - x$, $\forall\, x \ge 0$. The first and second order derivatives of $g(x)$ are given by $g'(x) = \alpha x^{\alpha-1} - 1$, $g''(x) = \alpha(\alpha - 1)x^{\alpha-2}$. As $\alpha \ge 1$ and $x \ge 0$, $g(x)$ is concave in $[0, +\infty)$, and its minimum is realized at $x^*_\alpha = \alpha^{\frac{1}{1-\alpha}}$. If $x^*_\alpha \ge \bar{x}_\alpha$, the minimum of $g(x)$, within the interval $[0, \bar{x}_\alpha]$, is given by $g(\bar{x}_\alpha) = 1 - [1 - (r - p)]^{1/\alpha} > 0$. That is $g(x) > 0$, $\forall\, x \in [0, \bar{x}_\alpha]$, which is equivalent to $f(x) > x$, $\forall\, x \in [0, \bar{x}_\alpha]$. Thus there exists no other equilibrium. If $x^*_\alpha < \bar{x}_\alpha$, the minimum of $g(x)$, within the interval $[0, \bar{x}_\alpha]$ is given by $g(x^*_\alpha) = r - p + \alpha^{\frac{\alpha}{1-\alpha}} - \alpha^{\frac{1}{1-\alpha}}$. We can verify that $g(x^*_\alpha)$ is decreasing in α, when $\alpha \ge 1$,

with $g(x^*_{\bar{\alpha}}) = 0$. Consequently, when $x^*_\alpha < \bar{x}_\alpha$ and $\alpha \le \bar{\alpha}$, $g(x) \ge 0$, $\forall\, x \in [0, \bar{x}_\alpha]$, and thus there exists no other equilibrium. To sum up, when $r - p < 1$, we identify two conditions, i.e., either $x^*_\alpha \ge \bar{x}_\alpha$, or $x^*_\alpha < \bar{x}_\alpha$ and $\alpha \le \bar{\alpha}$, under which there exists no other equilibrium. It is easy to verify that $x^*_\alpha \ge \bar{x}_\alpha$ is a sufficient condition for $\alpha \le \bar{\alpha}$, thus the two conditions are reduced to one inequality $\alpha \le \bar{\alpha}$.

Next consider the case when $r - p < 1$ and $\alpha > \bar{\alpha}$. As $g(x)$ is concave within the interval $[0, \bar{x}_\alpha]$, we know that there exist two points, say x_1 and x_2, such that $g(x_i) = 0$, $i = 1, 2$. Without loss of generality, we assume $x_1 < x_2$. As $g(0) = r - p > 0$, $g(\bar{x}_\alpha) = 1 - [1 - (r-p)]^{1/\alpha} > 0$ and the continuity of $g(x)$, $\forall\, x \in [0, \bar{x}_\alpha]$, we know that $g(x) > 0$, $\forall\, x \in [0, x_1) \cup (x_2, \bar{x}_\alpha]$, and $g(x) < 0$, $\forall\, x \in (x_1, x_2)$. Then we have $g'(x_1) < 0$ ($g'(x_2) > 0$), which is equivalent to $f'(x_1) < 1$ ($f'(x_2) > 1$). Thus, only the smaller root of $g(x) = 0$ constitutes another equilibrium, and we obtain the announced result. □

Proof of Proposition 3. Due to $\frac{1}{2}(x^\alpha - (1-x)^\alpha + 1) \in [0, 1]$, $\forall\, x \in [0, 1]$ and $\beta \in [0, 1]$, we can rewrite the updating probability function, as shown in Equation (7), as

$$f(x) = \frac{x^\alpha - (1-x)^\alpha + 1}{2} + \gamma \min\left\{1 - \frac{x^\alpha - (1-x)^\alpha + 1}{2}, \frac{(1-\beta)p}{2}\right\}.$$

Let us denote $f_1(x) = \frac{x^\alpha - (1-x)^\alpha + 1}{2} + \gamma\frac{(1-\beta)p}{2}$, and $f_2(x) = (1-\gamma)\frac{x^\alpha - (1-x)^\alpha + 1}{2} + \gamma$. When $(1-\beta)p > 2$, $\frac{(1-\beta)p}{2} > 1 - \frac{x^\alpha - (1-x)^\alpha + 1}{2}$, $\forall\, x \in [0, 1]$, and thus $f(x)$ is reduced to $f_2(x)$. When $(1-\beta)p \in [0, 2]$, there exists an $x' \in [0, 1]$, such that $f(x) = f_1(x), \forall\, x < x'$, and $f(x) = f_2(x), \forall\, x \ge x'$, where x' is the unique solution of $x^\alpha - (1-x)^\alpha = 1 - (1-\beta)p$. Notice that $(1-\beta)p \ge 1$ implies $x' \le 1/2$.

As the mapping $f : [0, 1] \to [0, 1]$ is monotone non-decreasing, the existence of the fixed points, i.e., $f(x) = x$, is guaranteed by Tarski's fixed point theorem (Tarski 1955). Next, we show that the fixed point is unique and stable when either condition is satisfied.

Consider first when $\alpha = 1$ or 2. Equation (7) is reduced to $f(x) = x + \gamma \min\left\{1 - x, \frac{(1-\beta)p}{2}\right\}$. The updating probability function $f(x)$ crosses the diagonal line only once at $x = 1$, when $1 - \beta > 0$, and $f'(1) = 1 - \gamma < 1$ for all $\gamma > 0$.

When $\alpha \in (1, 2)$, it is easy to verify that both $f_1(x)$ and $f_2(x)$ are convex when $x \in [0, \frac{1}{2}]$ and concave when $x \in [\frac{1}{2}, 1]$. $x = 1$ is guaranteed an equilibrium as $f(1) = 1$ and $f'(1) < 1$. To this end, we only need to show that there exists no fixed point in $[0, 1)$ when condition (i) is satisfied. Consider first when $(1-\beta)p \in [1, 2]$, i.e., $x' \le 1/2$. In this case, $f(x) = \min\{f_1(x), f_2(x\})$ for all $x \in [0, 1/2]$ and $f(x)$ is reduced to $f_2(x)$ for all $x \in [1/2, 1]$. Let $g_i(x) = f_i(x) - x$, $i = 1, 2$. As a linear combination of a concave function and a linear function, $g_2(x)$ is concave as well when $x \in [\frac{1}{2}, 1]$, and its minimum is reached at either $\frac{1}{2}$ or 1. We have $g_2(\frac{1}{2}) = \frac{\gamma}{2} > 0$ due to $\gamma > 0$, and $g_2(1) = 0$. Thus we conclude that $f(x)$ does not have a fixed point within $[\frac{1}{2}, 1)$.

Next we derive the conditions under which there exists no equilibrium in $[0,1/2]$. As $f_i(x)$ is convex within $[0,\frac{1}{2}]$, $g_i(x)$ is convex as well when $x\in[0,\frac{1}{2}]$. The minimum of $g_1(x)$ is reached at $g_1'(x)=0\Leftrightarrow x^{\alpha-1}+(1-x)^{\alpha-1}=\frac{2}{\alpha}$. Denote $\tilde{x}_1$ as the solution of $x^{\alpha-1}+(1-x)^{\alpha-1}=\frac{2}{\alpha}$. It is easy to verify that $x^{\alpha-1}+(1-x)^{\alpha-1}\in[1,2^{2-\alpha}]$ for all $x\in[0,\frac{1}{2}]$. As $2^{2-\alpha}>\frac{2}{\alpha}$ for all $\alpha\in(1,2)$, $\tilde{x}_1$ is within the region $[0,1/2]$. Next we show that under condition (i), $g_1(\tilde{x}_1)>0$. As $\tilde{x}_1\in[0,1/2]$, we have $\tilde{x}_1^\alpha+(1-\tilde{x}_1)^\alpha\le\tilde{x}_1^{\alpha-1}+(1-\tilde{x}_1)^{\alpha-1}=\frac{2}{\alpha}$, which is equivalent to $\tilde{x}_1^\alpha-(1-\tilde{x}_1)^\alpha\ge 2\tilde{x}_1^\alpha-\frac{2}{\alpha}$. Thus $g_1(\tilde{x}_1)\ge\tilde{x}_1^\alpha-\frac{1}{\alpha}+\frac{1}{2}+\gamma\frac{(1-\beta)p}{2}-\tilde{x}_1\ge\tilde{x}_1^2-\frac{1}{\alpha}+\frac{1}{2}+\gamma\frac{(1-\beta)p}{2}-\tilde{x}_1\ge\left(\tilde{x}_1-\frac{1}{2}\right)^2+\frac{1}{4}-\frac{1}{\alpha}+\gamma\frac{(1-\beta)p}{2}$. The second inequality is due to $\alpha\in(1,2)$. Thus under the condition that $\gamma(1-\beta)p\ge\frac{2}{\alpha}-\frac{1}{2}$, $g_1(\tilde{x}_1)$ is guaranteed to be greater than 0, and there exists no fixed point for $f_1(x)$ within $[0,1/2]$. Next we derive the conditions under which there exists no fixed point within $[0,1/2]$ for $f_2(x)$. The minimum of $g_2(x)$ is reached at $g_2'(x)=0\Leftrightarrow x^{\alpha-1}+(1-x)^{\alpha-1}=\frac{2}{(1-\gamma)\alpha}$. Let us denote the solution of $x^{\alpha-1}+(1-x)^{\alpha-1}=\frac{2}{(1-\gamma)\alpha}$ as $\tilde{x}_2$. So if $\tilde{x}_2$ is outside the interval $[0,\frac{1}{2}]$ (in this case, the minimum of $g_2(x)$ is reached at either 0 or $\frac{1}{2}$, which are both greater than 0), then $f_2(x)$ does not intersect with the diagonal line within $[0,\frac{1}{2}]$. As $x^{\alpha-1}+(1-x)^{\alpha-1}\in[1,2^{2-\alpha}]$ for all $x\in[0,\frac{1}{2}]$, one sufficient condition for $\tilde{x}_2\notin[0,\frac{1}{2}]$ is that $\frac{2}{(1-\gamma)\alpha}>2^{2-\alpha}$, which is equivalent to $\gamma>1-\frac{2^{\alpha-1}}{\alpha}$. When $(1-\beta)p<1$, $f(x)=f_1(x)$ for all $x\in[0,1/2]$, and $f(x)=\min\{f_1(x),f_2(x)\}$ for all $x\in[1/2,1]$. Thus, a unique stable equilibrium is guaranteed under the condition that $\gamma(1-\beta)p\ge\frac{2}{\alpha}-\frac{1}{2}$. Similarly, when $(1-\beta)p>2$, $f(x)$ is reduced to $f_2(x)$ for all $x\in[0,1]$, and the condition for a unique stable equilibrium is given by $\gamma>1-\frac{2^{\alpha-1}}{\alpha}$. Summarizing the conditions under the two scenarios, we thus obtain the announced result. □

Proof of Lemma 6. Given that the market is completely covered, the profit from producing the two products are given by $\pi_A(q_A)=pE[\min\{q_A,X^*\}]-c\cdot q_A-K$, $\pi_B(q_B)=pE[\min\{q_B,1-X^*\}]-c\cdot q_B-K$. Under the Impressionable Scenario, the demand in the limit X^* follows a Bernoulli distribution with support $\{0,1\}$, as shown in Lemma 5. Consider first when $n/T>1/2$. Suppose the optimal production quantity of product A is given by q_A^*. As $P(X^*=1)\le 1-g(T,k)$, the profit from producing q_A^* market share of product A satisfies $\pi_A(q_A^*)\le p[(1-g(T,k))\cdot q_A^*+g(T,k)\cdot 0]-c\cdot q_A^*-K=[p(1-g(T,k))-c]\cdot q_A^*-K$. As $q_A^*\in[0,1]$, the optimal profit from producing product A is no greater than 0 if $p(1-g(T,k))\le c+K$, and thus the firm would not produce product A. Similarly when $n/T\ge 1/2$ and $p(1-g(T,k))\le c+K$, the firm would obtain a non-positive profit from producing product B.

Under the Boundary Scenario, Lemma 5 shows that demand of product A, i.e., X^*, follows a $Beta(n,T-n)$ distribution. To streamline the proof, we first prove one intermediate result, and summarize it in the lemma below. Assume that X^* follows a general continuous probability

distribution $\Phi(\cdot)$ with density function $\phi(\cdot)$. With constant marginal cost c, the optimal production quantities of the two products are given by the following lemma.

LEMMA A2. *Given the constant marginal cost c, the optimal production quantities are given by $q_A^* = \Phi^{-1}\left(1-\frac{c}{p}\right)$ and $q_B^* = 1-\Phi^{-1}\left(\frac{c}{p}\right)$.*

Proof of Lemma A2. Given the assumption that X^* follows a continuous probability distribution $\Phi(\cdot)$ with density function $\phi(\cdot)$, the firm's profit function can be written as

$$\begin{aligned}\pi(q_A,q_B) = &-c(q_A+q_B)-2K+\min\{q_A+q_B,1\}\cdot p\int_{\min\{q_A,1-q_B\}}^{\max\{q_A,1-q_B\}}\phi(\xi)d\xi\\ &+p\int_{\max\{q_A,1-q_B\}}^{1}[q_A+(1-\xi)]\phi(\xi)d\xi+p\int_0^{\min\{q_A,1-q_B\}}(\xi+q_B)\phi(\xi)d\xi.\end{aligned}$$

When $q_A+q_B\geq 1$, the profit function is reduced to

$$\pi(q_A,q_B)=-c(q_A+q_B)-2K+p\int_{1-q_B}^{q_A}\phi(\xi)d\xi+p\int_{q_A}^{1}[q_A+(1-\xi)]\phi(\xi)d\xi+p\int_0^{1-q_B}(\xi+q_B)\phi(\xi)d\xi.$$

Its first order conditions (FOCs) are

$$\frac{\partial\pi}{\partial q_A}=-c+p\int_{q_A}^{1}\phi(\xi)d\xi,\qquad \frac{\partial\pi}{\partial q_B}=-c+p\int_0^{1-q_B}\phi(\xi)d\xi,$$

and the Hessian matrix is given by

$$H(\pi)=\begin{bmatrix}-p\phi(q_A) & 0\\ 0 & -p\phi(1-q_B)\end{bmatrix}.$$

The Hessian matrix is negative definite, and the optimal solutions are given by FOCs, which is $q_A^*=\Phi^{-1}(1-c/p)$ and $q_B^*=1-\Phi^{-1}(c/p)$. It is easy to verify that when $q_A+q_B<1$, FOCs and the Hessian matrix are the same as those when $q_A+q_B\geq 1$. We thus obtain the desired result. □

Given that X^* follows a $Beta(n,T-n)$ distribution, we have $q_A^*=I^{-1}_{1-c/p}(n,T-n)$ and $q_B^*=1-I^{-1}_{c/p}(n,T-n)$. Plugging q_A^* into the profit function from producing product A, we have

$$\begin{aligned}\pi_A(q_A^*) &= pE[\min\{q_A^*,X^*\}]-c\cdot q_A^*-K\\ &= p\left[\int_0^{q_A^*}x\cdot\frac{x^{n-1}(1-x)^{T-n-1}}{B(n,T-n)}dx+\int_{q_A^*}^{1}q_A^*\cdot\frac{x^{n-1}(1-x)^{T-n-1}}{B(n,T-n)}dx\right]-c\cdot q_A^*-K\\ &= \frac{p}{B(n,T-n)}\int_0^{q_A^*}x^n(1-x)^{T-n-1}dx-K\\ &= \frac{p\cdot n}{T}I_{q_A^*}(n+1,T-n)-K=\frac{p\cdot n}{T}I_{I^{-1}_{1-c/p}(n,T-n)}(n+1,T-n)-K.\end{aligned}$$

The firm would not produce product A if the optimal profit $\pi_A(q_A^*)$ is non-positive. Similarly, we can derive the optimal profit from producing product B, which is given by $\pi_B(q_B^*)=\frac{p\cdot n}{T}\left[1-I_{I^{-1}_{c/p}(n,T-n)}(n+1,T-n)\right]-K$. The firm would produce product B if and only if the optimal profit is positive, and thus we obtain the announced result. □

Proof of Proposition 4. Under the assumption of incomplete coverage without social influence, i.e., $0 < r - p < 1/2$, the random limits of the process can be divided into two categories, either the market is completely covered under social influence, or there exists no overlap in the market covered by the two products, i.e., each product acts like a local monopoly. Under the first case, the market equilibrium (x_A^*, x_B^*) is given by

$$\frac{1 + x_A^\alpha - x_B^\alpha}{2} = x_A, \qquad \frac{1 - x_A^\alpha + x_B^\alpha}{2} = x_B, \tag{9}$$

subject to the constraints $\frac{1+x_A^{*\alpha}-x_B^{*\alpha}}{2} \leq \Pi_{[0,1]}[r + x_A^{*\alpha} - p]$ and $\frac{1-x_A^{*\alpha}+x_B^{*\alpha}}{2} \leq \Pi_{[0,1]}[r + x_B^{*\alpha} - p]$. The solutions of equation set (9) are summarized in Lemma 3. Under the Boundary Scenario, the solution is $(x^*, 1-x^*)$, $\forall x^* \in [0,1]$. When $\alpha = 1$, plugging $(x^*, 1-x^*)$ into the constraints, we have $\frac{1+x^*-(1-x^*)}{2} = x^* \leq \Pi_{[0,1]}[r - p + x^*]$ and $\frac{1-x^*+(1-x^*)}{2} = 1 - x^* \leq \Pi_{[0,1]}[r - p + 1 - x^*]$. As $r - p > 0$, the constraints are satisfied for all $x^* \in [0,1]$, and thus the support of (X_A^*, X_B^*) is given by $\{(x^*, 1-x^*), \forall x^* \in [0,1]\}$. When $\alpha = 2$, the solutions of equation set (9) need to satisfy the following constraints: $\frac{1+x^{*2}-(1-x^*)^2}{2} = x^* \leq \Pi_{[0,1]}[r - p + x^{*2}]$ and $\frac{1-x^{*2}+(1-x^*)^2}{2} = 1 - x^* \leq \Pi_{[0,1]}[r - p + (1 - x^*)^2]$. The two inequalities are satisfied for all $x^* \in [0,1]$ when $r - p \geq 1/4$. Otherwise, the constraints are satisfied for $x^* \in \left[0, \frac{1-\sqrt{1-4(r-p)}}{2}\right] \cup \left[\frac{1+\sqrt{1-4(r-p)}}{2}, 1\right]$.

Under the Impressionable Scenario, Lemma 3 shows that the process converges to either $(0,1)$ or $(1,0)$. It is easy to verify that both $(0,1)$ and $(1,0)$ satisfy the constraints. Under the Obstinate Scenario, the process is shown to converge to $(1/2, 1/2)$. Plugging $(1/2, 1/2)$ into the first constraint, we have $\frac{1+x_A^{*\alpha}-x_B^{*\alpha}}{2} = \frac{1}{2} \leq \Pi_{[0,1]}\left[r - p + \left(\frac{1}{2}\right)^\alpha\right] \Leftrightarrow r - p \geq \frac{1}{2} - \left(\frac{1}{2}\right)^\alpha$. Plugging $(1/2, 1/2)$ into the second constraint yields the same condition. Consequently, $(1/2, 1/2)$ constitutes an equilibrium under the Obstinate Scenario if and only if $r - p \geq \frac{1}{2} - \left(\frac{1}{2}\right)^\alpha$.

Next consider the random limits when there exists no overlap in the coverage of the two products. In this case, random limit (x_A^*, x_B^*) is given by $\Pi_{[0,1]}[r + x_A^\alpha - p] = x_A, \Pi_{[0,1]}[r + x_B^\alpha - p] = x_B$, subject to the constraints $\Pi_{[0,1]}[r + x_A^{*\alpha} - p] < \frac{1+x_A^{*\alpha}-x_B^{*\alpha}}{2}$ and $\Pi_{[0,1]}[r + x_B^{*\alpha} - p] < \frac{1-x_A^{*\alpha}+x_B^{*\alpha}}{2}$. Similar to the proof of Lemma 4, we can show that (x_α^*, x_α^*) constitutes an equilibrium when $\alpha > \bar{\alpha}$, where x_α^* is the smaller root of $r + x^\alpha - p = x$, and $\bar{\alpha}$ is given by $\alpha^{\frac{1}{1-\alpha}} - \alpha^{\frac{\alpha}{1-\alpha}} = r - p$. That is $r + x^\alpha - p = x$ has two solutions within $[0,1]$ if and only if $\alpha > \bar{\alpha}$. Plugging (x_α^*, x_α^*) into the first constraint, we have $\Pi_{[0,1]}[r + x_\alpha^{*\alpha} - p] = x_\alpha^* < \frac{1+x_\alpha^{*\alpha}-x_\alpha^{*\alpha}}{2} = 1/2$. Let $g(x) = r + x^\alpha - p - x$. $x_\alpha^* < 1/2$ if and only if either $g(1/2) < 1/2$, or $g(1/2) \geq 1/2$ and $\arg\min_x g(x) < 1/2$. Notice that $g(1/2) = r + \left(\frac{1}{2}\right)^\alpha - p < 1/2 \Leftrightarrow r - p < \frac{1}{2} - \left(\frac{1}{2}\right)^\alpha$. It is easy to verify that $g(x)$ is convex for all $\alpha \geq 1$, and $\arg\min_x g(x) = \left(\frac{1}{\alpha}\right)^{\frac{1}{\alpha-1}}$. Thus we have that $\arg\min_x g(x) < 1/2$ is equivalent to $\alpha \in (1,2)$. As $\alpha^{\frac{1}{1-\alpha}} - \alpha^{\frac{\alpha}{1-\alpha}} \geq \frac{1}{2} - \left(\frac{1}{2}\right)^\alpha$, $\forall \alpha \geq 1$, (x_α^*, x_α^*) constitutes as an equilibrium if and only if either $r - p < \frac{1}{2} - \left(\frac{1}{2}\right)^\alpha$ and $\alpha \in \{1\} \cup [2, +\infty)$, or $r - p < \alpha^{\frac{1}{1-\alpha}} - \alpha^{\frac{\alpha}{1-\alpha}}$ and $\alpha \in (1,2)$. Thus, we obtain the announced result. □

Proof of Lemma 7. Let $\tilde{p} = 2r + x_A^\alpha + x_B^\alpha - 1 - p$, the highest price the firm can charge for product A such that a consumer is guaranteed to purchase either by product A or product B upon arrival. Let $p_{\min}^m = r + x_A^\alpha - 1$, the highest price the firm can charge for product A such that the product covers the entire market when only product A is available. Similarly, let $p_{\max}^m = r + x_A^\alpha$ be the lowest price to shut down demand for product A when only product A is available. Also let $p_{\min}^c = p + x_A^\alpha - x_B^\alpha - 1$, the highest the firm could charge for product A so that the entire market is covered when both products A and B are available. Let $p_{\max}^c = p + x_A^\alpha - x_B^\alpha + 1$, the lowest price to shut down demand for product A when both products are available.

First, let us find the optimal pricing without restraining the choice probabilities, i.e., $f_A(p_t, \boldsymbol{y})$ and $f_B(p_t, \boldsymbol{y})$, within the region $[0, 1]$. We will discuss the impact of this restriction on the optimal pricing afterwards. Without the restriction on $f_A(p_t, \boldsymbol{y})$ and $f_B(p_t, \boldsymbol{y})$, $g_t(p_t, \boldsymbol{y})$ is reduced to

$$g_t(p_t, \boldsymbol{y}) = \begin{cases} \frac{\left(x_A^\alpha - x_B^\alpha + 1 + p - p_t\right)}{2}(p_t - \Delta_A V_{t-1}(\boldsymbol{y})) + \frac{\left(x_B^\alpha - x_A^\alpha + 1 - p + p_t\right)}{2}(p - \Delta_B V_{t-1}(\boldsymbol{y})) & \text{if } p_t \le \tilde{p}, \\ (r - p_t + x_A^\alpha)(p_t - \Delta_A V_{t-1}(\boldsymbol{y})) + (r - p + x_B^\alpha)(p - \Delta_B V_{t-1}(\boldsymbol{y})) & \text{if } p_t > \tilde{p}. \end{cases}$$

The first order and the second order conditions with respect to p_t are given by, when $p_t \le \tilde{p}$, $g_t'(p_t, \boldsymbol{y}) = \frac{1}{2}\Big(x_A^\alpha - x_B^\alpha + 1 + p - p_t\Big) - \frac{1}{2}\Big(p_t - \Delta_A V_{t-1}(\boldsymbol{y})\Big) + \frac{1}{2}\Big(p - \Delta_B V_{t-1}(\boldsymbol{y})\Big)$ and $g_t''(p_t, \boldsymbol{y}) = -1$. Consequently, $g_t(p_t, \boldsymbol{y})$ is concave in p_t when $p_t \le \tilde{p}$, and we can verify that its maximum is realized at $p_c^* = p + \frac{1}{2}\Big(\Delta_A V_{t-1}(\boldsymbol{y}) - \Delta_B V_{t-1}(\boldsymbol{y}) + x_A^\alpha - x_B^\alpha + 1\Big)$, if $p_c^* \le \tilde{p}$. Otherwise, the maximum of $g_t(p_t, \boldsymbol{y})$, $\forall\, p_t \le \tilde{p}$ is given by $g_t(\tilde{p}, \boldsymbol{y})$. Similarly, when $p_t > \tilde{p}$, the first order and the second order conditions are given by $g_t'(p_t, \boldsymbol{y}) = -(p_t - \Delta_A V_{t-1}(\boldsymbol{y})) + (r - p_t + x_A^\alpha)$ and $g_t''(p_t, \boldsymbol{y}) = -2$. Thus, $g_t(p_t, \boldsymbol{y})$ is concave in p_t as well when $p_t > \tilde{p}$, and its maximum is realized at $p_m^* = \frac{1}{2}\Big(\Delta_A V_{t-1}(\boldsymbol{y}) + x_A^\alpha + r\Big)$, if $p_m^* > \tilde{p}$. Otherwise, the maximum of $g_t(p_t, \boldsymbol{y})$, $\forall\, p_t > \tilde{p}$ is given by $g_t(\tilde{p}, \boldsymbol{y})$.

Next let us discuss the impact of restraining the choice probabilities $f_A(p_t, \boldsymbol{y})$ and $f_B(p_t, \boldsymbol{y})$ within $[0, 1]$ on the optimal pricing. When $r - p + x_B^\alpha \ge 1$, i.e., $x_B \ge (1 - r + p)^{\frac{1}{\alpha}}$, the choice probabilities are reduced to $f_A(p_t, \boldsymbol{y}) = \Pi_{[0,1]}\left[\frac{x_A^\alpha - x_B^\alpha + 1 + (p - p_t)}{2}\right]$, and $f_B(p_t, \boldsymbol{y}) = \Pi_{[0,1]}\left[\frac{x_B^\alpha - x_A^\alpha + 1 - (p - p_t)}{2}\right]$. That is the market is completely covered no matter what p_t is. Recall that we defined $p_{\min}^c = p + x_A^\alpha - x_B^\alpha - 1$ and $p_{\max}^c = p + x_A^\alpha - x_B^\alpha + 1$. We can verify that $p_{\min}^c$ is the largest p_t such that $f_A(p_t, \boldsymbol{y}) = 1$ and $f_B(p_t, \boldsymbol{y}) = 0$, and $p_{\max}^c$ is the smallest p_t such that $f_A(p_t, \boldsymbol{y}) = 0$ and $f_B(p_t, \boldsymbol{y}) = 1$. As $g_t(p_t, \boldsymbol{y})$ is concave within $[p_{\min}^c, p_{\max}^c]$, and $g_t(p_t, \boldsymbol{y}) < g_t(p_{\min}^c, \boldsymbol{y})$, $\forall\, p_t < p_{\min}^c$, and $g_t(p_t, \boldsymbol{y}) = g_t(p_{\max}^c, \boldsymbol{y})$, $\forall\, p_t > p_{\max}^c$, we can restrict the search of optimal p_t within the region $[p_{\min}^c, p_{\max}^c]$. If $p_c^* \in [p_{\min}^c, p_{\max}^c]$, then $g_t(p_t, \boldsymbol{y})$ is maximized at p_c^*. Otherwise, the maximum of $g_t(p_t, \boldsymbol{y})$ is realized at either $p_{\min}^c$ if $p_c^* < p_{\min}^c$ or $p_{\max}^c$ if $p_c^* > p_{\max}^c$.

When $r-p+x_B^\alpha<1$, i.e., $x_B<(1-r+p)^{\frac{1}{\alpha}}$, the choice probabilities $f_A(p_t,\boldsymbol{y})$ and $f_B(p_t,\boldsymbol{y})$ are given by

$$f_A(p_t,\boldsymbol{y})=\begin{cases}1 & \text{if } p_t\le p^c_{\min},\\ \frac{1}{2}\left(x_A^\alpha-x_B^\alpha+1+p-p_t\right) & \text{if } p_t\in(p^c_{\min},\tilde{p}],\\ r-p_t+x_A^\alpha & \text{if } p_t\in(\tilde{p},p^m_{\max}],\\ 0 & \text{if } p_t>p^m_{\max},\end{cases}$$

and

$$f_B(p_t,\boldsymbol{y})=\begin{cases}0 & \text{if } p_t\le p^c_{\min},\\ \frac{1}{2}\left(x_B^\alpha-x_A^\alpha+1-p+p_t\right) & \text{if } p_t\in(p^c_{\min},\tilde{p}],\\ r-p+x_B^\alpha & \text{if } p_t\in(\tilde{p},p^m_{\max}],\\ 1 & \text{if } p_t>p^m_{\max}.\end{cases}$$

As shown in the above analysis, $g_t(p_t,\boldsymbol{y})$ is concave in p_t within $[p^c_{\min},\tilde{p}]$, and within $[\tilde{p},p^m_{\max}]$. As $g_t(p_t,\boldsymbol{y})\le g_t(p^c_{\min},\boldsymbol{y})$, $\forall\, p_t\le p^c_{\min}$, and $g_t(p_t,\boldsymbol{y})=g_t(p^m_{\max},\boldsymbol{y})$, $\forall\, p_t>p^m_{\max}$, we can restrict the search of optimal price p_t within the region $[p^c_{\min},p^m_{\max}]$. In order to find the optimal pricing, we need to verify whether p_c^* and p_m^* fall within their respective regions $[p^c_{\min},\tilde{p}]$ and $[\tilde{p},p^m_{\max}]$. In total we have nine cases, and the results are summarized in the lemma. □

Proof of Lemma A1. When consumers are susceptible to others' decisions, the sales process is characterized by Equation (7). The convergence rate of the process can be derived from Arthur et al. (1987, Theorems 2 and 3), i.e.,

$$\limsup_{t\to\infty}\sqrt{\frac{t}{\ln\ln t}}|X_t-x^*|=\begin{cases}\sqrt{\frac{2x^*(1-x^*)}{1-2f'(x^*)}} & \text{if } f'(x^*)<\frac{1}{2},\\ +\infty & \text{if } f'(x^*)\ge\frac{1}{2}.\end{cases}\tag{10}$$

Compare it with the benchmark case where all consumers' purchase decisions are solely based upon their own preferences. In this case, any random consumer purchases product A with probability $y^*\equiv(1-\gamma)/2+\gamma\Pi_{[0,1]}\left[\frac{1+(1-\beta)p}{2}\right]$, and purchase product B with probability $1-y^*$. Denote, by Y_t, the fraction of consumers who purchase product A at the beginning of period t. The convergence rate of the process $\{Y_t\}$ is given by (Durrett 2010, Section 2.5.1 and Theorem 8.8.2),

$$\limsup_{t\to\infty}\sqrt{\frac{t}{\ln\ln t}}|Y_t-y^*|=\sqrt{2y^*(1-y^*)}.\tag{11}$$

Notice that $f(\frac{1}{2})=y^*$, and thus we obtain the desired result by comparing the right-hand-sides of Equations (10) and (11). □

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